

Rock physics and petrophysical evaluation of diagenetic effects in the Broom Creek CO₂ sequestration site

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ABSTRACT

The Pennsylvanian–Permian Broom Creek Formation, a deep saline aquifer in the Williston Basin, North Dakota, is an important target for large-scale CO₂ sequestration. Preserving long-term CO₂ storage integrity in this reservoir hinges on a systematic assessment of how geological and diagenetic processes modify rock properties. This study investigates the impact of diagenetic alterations on the elastic properties of the Broom Creek sandstone through an integrated rock physics modeling approach. Multi-mineral petrophysical evaluation was conducted using geophysical well logs calibrated with X-ray diffraction and cuttings data. Rock physics models, including the contact cement, soft-sand, and stiff-sand models, were employed to establish diagenetic trends in compressional wave velocity and porosity space. Velocity-porosity trends and mineral volume fractions data were then used to quantify the contributions of contact cement and pore-filling materials to the rock framework. The findings reveal Upper Broom Creek eolian quartz arenite deposits exhibit diagenetic contact cementation by quartz overgrowths and dolomite, which stiffen the rock matrix while preserving porosity and permeability. In the lower, near-shore sandstone intervals of the formation, quartz overgrowths and dolomite cementation persist, but pore-filling anhydrite and clay dominate as non-contact cements. These results provide understanding into the mechanical properties of the reservoir and the role of diagenesis in modifying porosity and stiffness. The study's outcomes are useful for modeling CO₂-brine-rock interactions and evaluating the long-term stability and effectiveness of CO₂ sequestration in the Broom Creek Formation.

1. Introduction

Carbon dioxide emissions, mainly due to anthropogenic activities, are one of the primary drivers of global warming (Pachauri et al., 2014; Al-Ghussain, 2019). The Intergovernmental Panel on Climate Change (IPCC) emphasizes the urgency of reducing CO₂ emissions to limit global warming to 1.5 °C, requiring significant efforts in carbon capture and storage (CCS) technologies alongside emission reductions (Metz et al., 2005; IPCC, 2022). Geological storage of CO₂, particularly in deep saline aquifers, plays a primary role in achieving these targets by providing a long-term and secure means of isolating CO₂ from the atmosphere (Lal, 2008; Benson and Cole, 2008). Deep saline aquifers are especially important due to their widespread availability, large storage capacity, and suitability for sequestration at industrial scales (Bachu, 2015; Kapetaki et al., 2016; Luo et al., 2022b).

The Broom Creek Formation within the Williston Basin in North Dakota is an important subsurface formation for CO₂ sequestration, exemplified by large-scale commercial projects such as the Red Trail Energy CCS project and Project Tundra (Leroux et al., 2020, 2022; Peck

et al., 2024). These projects leverage the formation's unique geological characteristics to store CO₂ underground safely. The Red Trail Energy project, near Richardton, injects about 180,000 tons of CO₂ annually into the Broom Creek Formation (Leroux et al., 2020; UNDEERC, 2024). Project Tundra, managed by Minnkota Power Cooperative, aims to capture and geologically store up to 4 million metric tons of CO₂ annually from the Milton R. Young Station, a lignite coal-fired power plant (U.S. Department of Energy, 2022). The project targets both the Broom Creek and Deadwood Formations, with regulatory approval from the North Dakota Industrial Commission, making it one of the largest CO₂ capture initiatives globally (Peck et al., 2024; NDIC, 2024b). Geological studies of the Broom Creek Formation, including core samples and geophysical surveys, carried out by the University of North Dakota Energy and Environmental Research Center (EERC) and partners have facilitated the creation of detailed 3D models predicting CO₂ movement and ensuring secure storage (Peck et al., 2019). The EERC continues to enhance monitoring technologies to verify CO₂ containment and reduce long-term monitoring costs (Barajas-Olalde et al., 2022, 2023b).

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Continuous monitoring of CO₂ injection sites and reservoir stability is imperative because it ensures the effective long-term storage of CO₂, minimizes the risk of leakage, and maintains the integrity of the storage reservoir (Ringrose et al., 2013). CO₂ injection induces alterations in both rock and fluid properties within the sequestration system. When CO₂ interacts with brine, it dissolves into the solution, forming carbonic acid, which then partially dissociates into hydrogen ions and bicarbonate ions, causing the pH of the brine to decrease (Berrezueta et al., 2023; Chen et al., 2023). Rock properties change, such as mineral and cement dissolution, cement precipitation, precipitation of pore-filling materials, due to chemical reactions, change in pore pressure, and alterations in fracture properties due to mechanical and petrophysical changes (Khan et al., 2024; Peter et al., 2022; Prasad et al., 2021).

Understanding the mineralogy of the storage formation is important for predicting and managing the chemical and mechanical alterations induced by CO₂ injection (Marbler et al., 2013). Particularly those acting as grain-contact cements and pore-filling materials play a key role in determining the extent of these alterations, with depositional settings and diagenesis being controlling factors influencing the cementation processes in sandstone reservoirs. Grain contact cement, such as quartz overgrowth, calcite, and dolomite cementation are crucial to maintaining the mechanical stability of the rock framework; Their dissolution could weaken the mechanical strength of the formation and reduce the integrity of the reservoir (Kim et al., 2018). However, the reduction in mechanical integrity is also influenced by the cementation type and compaction history and depth, with carbonate-cemented sandstone undergoes substantial changes among sedimentary rocks (Wu et al., 2023; Wang et al., 2023). Conversely, pore-filling materials, such as clays, strongly influence porosity and permeability. Their dissolution or precipitation during CO₂ injection can either enhance or obstruct fluid flow pathways, significantly affecting storage capacity and injectivity (Kim et al., 2018; Zhang et al., 2024; Gao et al., 2024). Understanding the diagenetic evolution of these materials is therefore essential for predicting their impact on reservoir quality during CO₂ storage. Assessing the mineralogical composition and its spatial distribution is therefore essential for modeling the behavior of the storage formation and assessing the long-term stability of CO₂ sequestration.

Building on this understanding, the objective of the present study is to investigate how diagenetic alterations affect the rock's elastic properties Broom Creek storage reservoir. Specifically, this research aims to identify and quantify the grain-contact cements and pore-filling materials within the reservoir layers of the Broom Creek Formation using combined analysis of multi-mineral petrophysical evaluation, rock physics modeling and petrographic analysis. Addressing these objectives will provide important observations for reservoir monitoring during long-term CO₂ storage, particularly as rock-fluid interactions alter the mechanical and petrophysical properties of the storage formation.

The paper is organized as follows: Section 2 provides an overview of the geological background of the Broom Creek Formation. Section 3 describes the data availability, along with the methodologies employed for petrophysical analysis and rock physics modeling. Section 4 presents a systematic discussion of the results, including key findings from the petrophysical evaluation, thin section analysis, and insights from rock physics modeling. Finally, Section 5 summarizes the conclusions of this study.

2. Geological background and study area

The Williston Basin is an intracratonic sedimentary basin spanning North Dakota, South Dakota, Montana, Saskatchewan, and Manitoba (refer to Fig. 1). It contains sedimentary deposits ranging from Cambrian to Tertiary ages (Gerhard, 1982). Within the Williston basin, the Broom Creek Formation, part of the upper Minnelusa Group of Permian age (Rygh, 1990), is a brine-saturated unit characterized by highly

Table 1
Summary of geophysical well logs, core analysis, and petrographic analysis used in the present study.

Well logs	Core analysis	Petrographic analysis
GR	Routine Core Analysis (Porosity & Permeability)	Mineralogy
RT		Grain size and texture
RHOZ		Cement types and distribution
NPHI	XRD Analysis (Mineralogy)	Diagenetic features
PEF		
DTC		
DTS		
K		

porous and permeable sandstone interspersed with dolomite and anhydrite providing significant pore volume for storage (Sorensen et al., 2009; Pothana et al., 2024) (see Fig. 2). It is further distinguished by an excellent sealing system, with the low-permeability Opeche Formation serving as an overlying caprock and the Amsden Formation as a basal barrier (Peck et al., 2019) (see Fig. 2). These attributes make the Broom Creek Formation a prime candidate for geological CO₂ storage (Peck et al., 2019; Sorensen et al., 2009).

The Broom Creek Formation represents a complex depositional environment predominantly influenced by eolian processes intercalated with nearshore marine and carbonate settings (Rygh, 1990). Eolian sandstone facies, characterized by cross-bedding and steeply inclined laminae, dominate the formation and reflect high-energy depositional conditions, while nearshore marine and carbonate facies indicate intermittent marine transgressions and regressions (Rygh, 1990; Operator, 2024). Diagenesis in the Broom Creek Formation has significantly impacted its reservoir quality, with processes such as dolomitization, quartz overgrowths, and localized precipitation of anhydrite and gypsum cementing pore spaces (Operator, 2024).

3. Materials and methods

3.1. Data

The dataset comprises geophysical well logs, conventional core analysis results, XRD analysis results, and petrographic images with their interpretations, as presented in Table 1. The data belong to five wells named Well-1, Well-2, Well-3, Well-4, and Well-5 for the present study. The well log and core data were acquired between 2018 to 2023 as part of regulatory submissions to the North Dakota Industrial Commission (NDIC). These data sets are sourced from the Department of Mineral Resources, North Dakota Industrial Commission (NDIC, 2024a). The geophysical well logs include gamma ray (GR), formation resistivity (RT), bulk density (RHOZ), neutron porosity (NPHI), photoelectric factor logs (PEF), compressional wave slowness (DTCO), shear wave slowness (DTS), and natural gamma spectroscopy logs (NGS). Core analysis data consists of routine measurements of porosity and permeability, as well as X-ray diffraction (XRD) results. Petrographic data consist of thin-section images accompanied by detailed interpretations of mineralogy and diagenetic features. The study focuses on five wells located in Oliver County and Stark County, North Dakota, as depicted in Fig. 1, situated in the green-boxed area in the Williston Basin.

3.2. Workflow

The workflow for this study is presented in Fig. 3, outlining the key steps undertaken. The process begins with the quality control of geophysical well logs to ensure reliable input data. This is followed by a multi-mineral petrophysical evaluation, validated against XRD mineral analysis and cuttings data. Next, thin section analysis results are synthesized to provide understanding into mineralogy and diagenetic features. A rock physics template is then constructed, enabling the computation of grain-contact cements and pore-filling cements, which are further

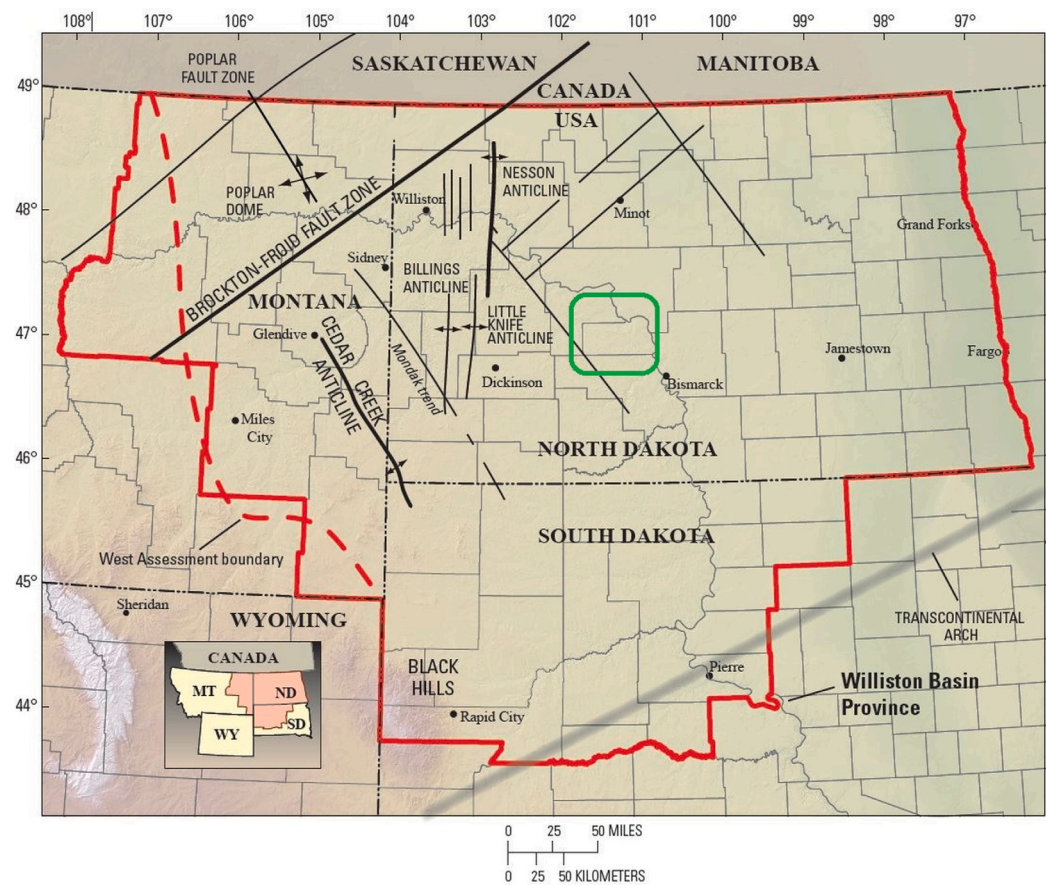


Fig. 1. Study Area: Geological map of the Williston Basin in North Dakota, showcasing its key geological features. The study specifically emphasizes area marked in green.
Source: Adapted from Gaswirth et al. (2013).

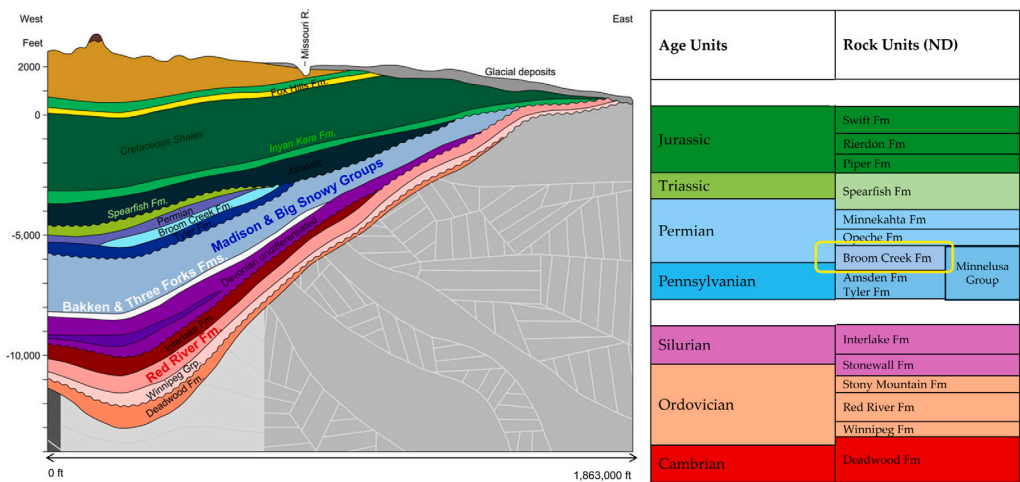


Fig. 2. Generalized stratigraphy of the Williston Basin illustrating selected geological age units: the left panel shows a west-east cross-section with a vertical exaggeration of approximately 133 ×, while the right panel presents a stratigraphic column.
Source: Figure adapted from Barajas-Olalde et al. (2023a), with base graphic courtesy of the North Dakota Geological Survey (NDGS, 2025).

validated using mineralogical data and thin section analysis. Finally, an combined interpretation is made, linking the diagenetic alterations to variations in rock stiffness, synthesizing the findings from all analytical steps.

3.2.1. Log quality control

Geophysical well logs consist of discrete measurements taken along the wellbore at regular intervals, known as the sampling interval. When geophysical measurements are recorded during different log-

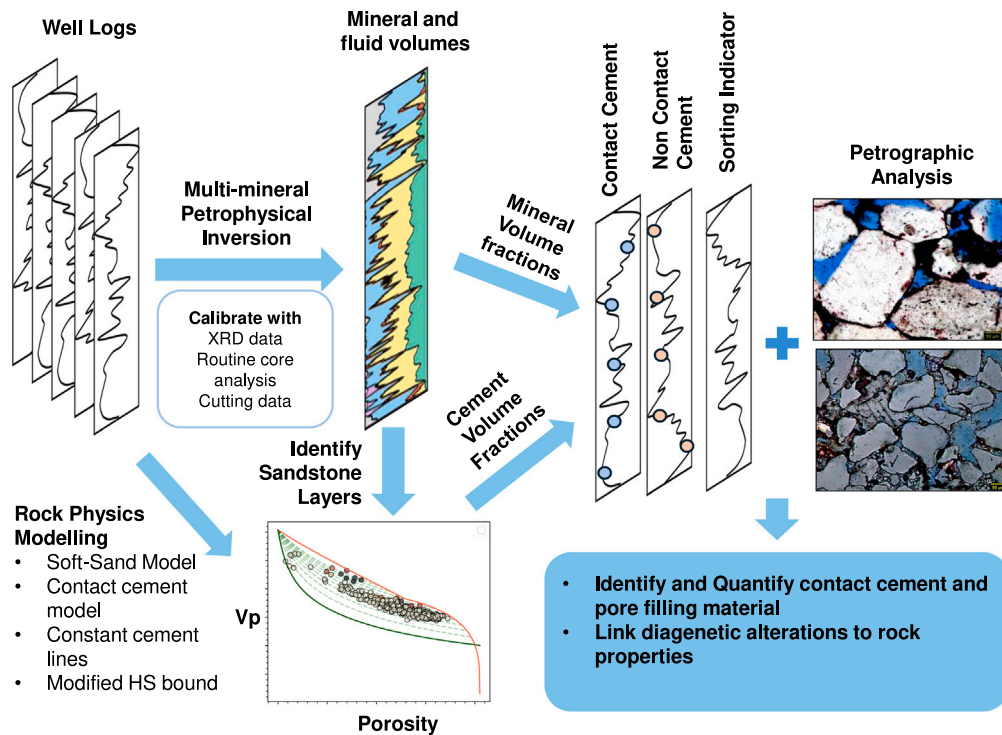


Fig. 3. Workflow illustrating the integrated analysis process utilized in this study. The methodology incorporates well-log data, multi-mineral petrophysical inversion calibrated with XRD data, routine core analysis, and cuttings data, rock physics modeling and integration of results from thin section analysis and rock physics modeling.

ging runs, there is a potential for depth mismatches between these datasets. To ensure consistency, it is crucial to align all geophysical measurements to a common depth reference. In this study, the gamma-ray (GR) log was used as the reference curve, and all other well logs were depth-matched accordingly. Following depth correction, the next step is to assess borehole conditions and their influence on logging measurements. Logging tools with shallow depths of investigation, such as pad-type tools, are particularly sensitive to borehole washouts and rugosity. Therefore, evaluating the extent to which these logs are affected is a necessary step. The borehole caliper log serves as a valuable tool for this assessment. We conducted a thorough analysis of the borehole conditions for each studied well in the Broom Creek Formation and excluded wells with significant borehole deterioration. Although it is possible to correct for borehole washout effects, we avoided this approach in this study due to its inherent subjectivity and the residual uncertainty in the corrected values. Additionally, for meaningful comparisons, accurate depth referencing between routine and special core analyses and wireline logs is essential. In the wells studied, core gamma-ray logs were available, and we ensured that the core gamma-ray data aligned with the wireline gamma-ray logs, confirming the accuracy of the depth standardization.

3.3. Petrophysical evaluation

The goal of multi-mineral petrophysical inversion is to estimate both mineral and fluid volumes from geophysical well logs. This process involves several key steps: (i) identifying the relevant input logs, (ii) selecting the target minerals and fluids, (iii) formulating the geophysical inverse problem, which includes choosing suitable saturation models, (iv) solving the inversion problem through an iterative approach, and (v) determining the best-fit model that aligns closely with core data, drilling cuttings, and prior geological knowledge. Multi-mineral petrophysical analysis is a well-established technique commonly employed in the industry to estimate mineral volumes and petrophysical parameters. For an in-depth theoretical explanation of the inverse problem and the mineral volume estimation process, refer to [Doveton \(1994\)](#).

Table 2

List of input logs and target minerals used in the multi-mineral petrophysical analysis.

Well logs	Minerals & fluids
GR	Clay
RT	Anhydrite
RHOZ	Dolomite
NPHI	K-Feldspar
DTCO	Quartz
PEF	Brine
K	

The geophysical well logs include gamma ray (GR), formation resistivity (RT), bulk density (RHOZ), neutron porosity (NPHI), photo-electric factor logs (PEF), compressional wave slowness (DTCO), and potassium concentration (K) from natural gamma spectroscopy logs were selected for the petrophysical analysis. Based on XRD analysis, cross-plot analysis, and cuttings data, the dominant minerals in the formation were identified as quartz, dolomite, feldspar, anhydrite, and clay (e.g., see [Fig. 4](#)). To simplify the inverse problem and avoid an underdetermined formulation, illite, along with minor quantities of kaolinite and chlorite, are grouped as “clay” in this study. This is also due to the relatively low clay content in the Broom Creek Formation (e.g., see [Fig. 4](#)). It is important to note that it is practically impossible to determine all the minerals present in the formation; therefore, only the dominant minerals were selected in the present study. The table below summarizes the input logs and target minerals (see [Table 2](#)). Since the Broom Creek Formation is brine-saturated, only a single fluid phase (brine) is considered, with the assumption that the entire pore space is filled with brine.

The forward problem is formulated as a set of linear equations, where each equation represents a linear combination of the mineral

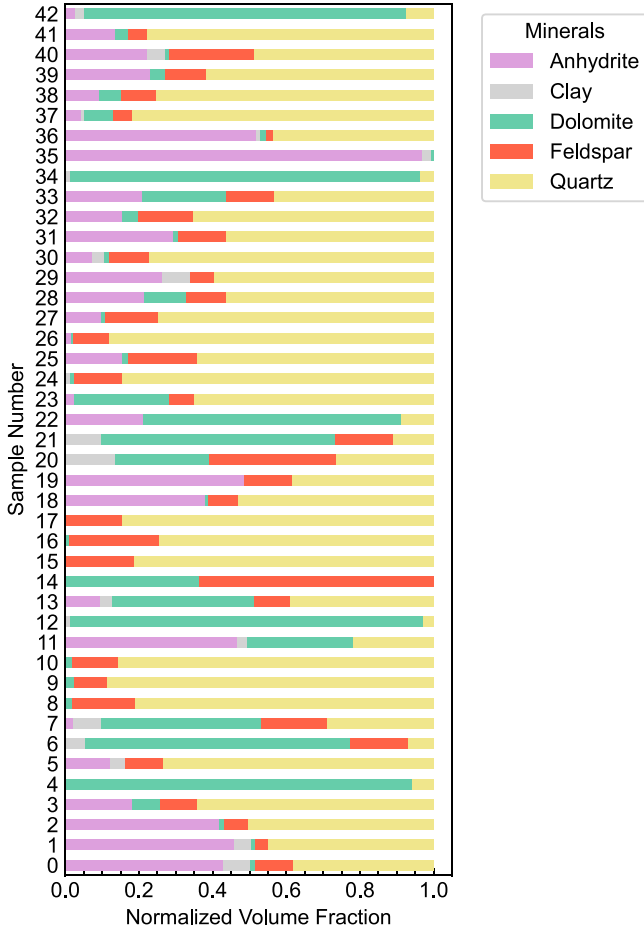


Fig. 4. XRD data analysis results for samples from the well-2. The plot shows the normalized volume fractions of key minerals, including anhydrite, clay, dolomite, feldspar, and quartz, across multiple samples.

volume fractions in the formation and the corresponding log measurements. Mathematically, this can be expressed as:

$$y_i = \sum_{j=1}^N R_j x_j + n \quad (1)$$

where y_i is the measured log value, x_i is the volume fraction of the i th mineral or fluid constituent, and R_i is the corresponding response of the i th constituent to the log measurement. N represents the total number of minerals and fluids being solved for, and n is the noise term accounting for measurement errors and uncertainties. In multi-mineral analysis, the endpoints of the response functions are iteratively adjusted within the reasonable bounds, and the log response is predicted. The iterative process continues until the error between the predicted value and the measured value (y_p) is less than the specified error tolerance, expressed as $|y_i - y_p| \leq \delta$. It is important to note that the best mathematical solution may not always accurately represent the geological formation. Therefore, validation with external data — such as X-ray diffraction (XRD) results, drill cuttings analysis, and existing geological knowledge — provides governing constraints. These validations help determine when to stop the iterative process and ensure that the model correctly reflects the subsurface conditions. Petrophysical inversion was performed using Schlumberger's Techlog (Schlumberger, 2022), locally installed on the University of North Dakota's Computational Research Center server. Furthermore, since the Broom Creek Formation sandstone intervals contains minimal amounts of clay minerals, we opted to use Archie's equation instead of more complex shaly sand models (Archie, 1942). The following Archie's parameters were applied:

$S_w = 1$, $a = 1$, $m = 1.8$, $n = 2$, and $R_w = 0.05 \Omega m$ at 150 °F (Pothana et al., 2024). Owing to the exclusively brine-saturated nature of this formation, Archie's equation effectively constrains porosity estimation via the resistivity log rather than fluid-saturation estimations.

3.4. Rock physics modeling

High-porosity sandstones are often modeled using contact models due to their simplicity and versatility. These models idealize the rock as an aggregate of spherical grains, each with specific mechanical properties. The effective elastic properties of these granular rocks at any given depth can be estimated using a combination of mineral elastic properties, the initial packing of the grains, and the effective pressure at burial depth.

Initially, during sedimentation, grains are suspended in water or air, and their porosity is largely controlled by sorting and grain shape. Once these suspensions being deposited, well-sorted clean sandstones typically exhibit initial porosities around 0.4–0.45 (Giles, 1997; Ajdukiewicz and Lander, 2010). This early stage is especially important, as the initial packing of grains defines the coordination number (n) — the average number of contacts per grain — and the initial porosity before diagenesis, which represents the porosity of freshly deposited sediments. At this stage, the effective elastic properties of the rock are close to the Reuss bound, the softest configuration of grain packing, reflecting minimal grain interaction (Mavko et al., 2020a).

As burial progresses, mechanical compaction, cementation, and other diagenetic alterations decrease the porosity of the rock. Diagenesis in sandstones can be categorized into three stages, each with distinct processes (Worden and Burley, 2003). Eogenesis, early diagenesis, involves all the processes that occur at the surface, including bioturbation, organic matter oxidation, and the formation of iron oxides. Mesogenesis, burial diagenesis, is dominated by compaction, pressure solution, cementation, and thermal maturation as sediments are subjected to increased temperature and pressure. Telogenesis, late diagenesis, occurs during uplift and exhumation, characterized by fracture filling, retrograde reactions, and alteration by meteoric water (Worden and Burley, 2003). These stages collectively shape the physical and chemical properties of sedimentary rocks. These processes simultaneously increase the rock's effective elastic properties, driving them from the Reuss lower bound towards the elastic properties of the minerals themselves. However, the rate and nature of this progression depend on the specific diagenetic alterations the rock undergoes. During mesogenesis, mechanical compaction primarily reduces porosity by rearranging grains, while cementation strengthens grain contacts and enhances rock stiffness. Mineral precipitation in pore spaces also contributes to increased stiffness and higher seismic velocities, though its impact is generally less pronounced than that of cementation at grain contacts.

The first step in modeling granular rocks is to compute the end-points: the mineral elastic properties at theoretical zero porosity and the effective elastic properties at critical porosity (ϕ_c) (see step-1 in Fig. 5). The elastic properties of minerals at zero porosity define the upper bound, whereas those at critical porosity represent the lower bound of rock stiffness at the given burial depth. The critical porosity represents the porosity at the hydrostatic pressure, a starting point for understanding how cementation influences porosity and rock stiffness. The mineral elastic moduli can be estimated using Hill's average, which provides a means to compute an effective elastic property by averaging between the Voigt and Reuss bounds, expressed as (e.g., Mavko et al., 2020b)

$$K_H = \frac{1}{2} \left[\sum_i f_i K_i + \frac{1}{\sum_i \frac{f_i}{K_i}} \right] \quad (2)$$

$$G_H = \frac{1}{2} \left[\sum_i f_i G_i + \frac{1}{\sum_i \frac{f_i}{G_i}} \right] \quad (3)$$

Table 3

Elastic and physical properties of input minerals, critical porosity and coordination number used in the rock physics modeling (Mavko et al., 2020c).

Component	Bulk modulus (K, GPa)	Shear modulus (G, GPa)	Density (ρ , g/cm ³)
Anhydrite	62.1	33.6	2.98
K-Feldspar	37.5	15.0	2.62
Quartz	36.6	45.0	2.65
Dolomite	94.0	45.0	2.87
Clay	11.5	15.3	2.62
Brine	2.3	0	1.03
Model Parameters	Value	Unit	Notes
Critical porosity (ϕ_0)	0.40	(v/v)	
Coordination number	8	–	

Where, f_i is the volume fraction of the i th constituent, K_i is the bulk modulus of the i th constituent, G_i is the shear modulus of the i th constituent. The elastic properties of minerals used in the present study are summarized in Table 3.

The effective elastic properties at the critical porosity can be estimated using the Hertz–Mindlin (HM) contact theory (e.g., Mavko et al., 2020b)

$$K_{HM} = \left(\frac{n^2(1-\phi_c)^2 G^2}{18\pi^2(1-\nu)^2} P \right)^{\frac{1}{3}} \quad (4)$$

$$G_{HM} = \frac{2+3f-\nu(1+3f)}{5(2-\nu)} \left(\frac{3n^2(1-\phi_c)^2 G^2}{2\pi^2(1-\nu)^2} P \right)^{\frac{1}{3}} \quad (5)$$

The variable K_{HM} and G_{HM} represents the effective bulk modulus and effective shear modulus of the granular medium as predicted by Hertz–Mindlin contact theory. The variable ϕ_c is the porosity at which the granular material transitions from a load-bearing to a non-load-bearing structure. G represents the shear modulus of the individual grains, and ν is Poisson's ratio of the individual grains. P refers to the effective pressure applied to the granular medium. f represents the fraction of grain contacts exhibiting complete adhesion. When $f = 0$, the grains are entirely frictionless, while $f = 1$ indicates perfect adhesion at all grain contacts. n is the coordination number. The coordination number can be approximated using the formula, which relates the coordination number (n) and porosity (ϕ_c) of sandstone rock (Murphy, 1982; Avseth et al., 2010).

$$n = 20 - 34\phi_c + 14\phi_c^2 \quad (6)$$

Using this formula, we have used the coordination number 8 for the critical porosity 0.4 (see Table 3).

Using these two endpoints — the pure mineral elastic properties at zero porosity and the critical porosity from the HM model —, we can apply the Hashin–Shtrikman (HS) lower and upper bounds to estimate the theoretical elastic properties at varying intermediate porosities below the critical porosity. The HS lower bound represents the softest possible mineral rearrangement, where the grains interact minimally, leading to limited stiffening of the rock (see step-2 in Fig. 5). This behavior is modeled by what is commonly known as the friable sand or soft sand model, which can be expressed as (Dvorkin and Nur, 1996):

$$K_{dry} = \left(\frac{\frac{\phi}{\phi_c}}{K_{HM} + \frac{4G_{HM}}{3}} + \frac{1-\frac{\phi}{\phi_c}}{K_{min} + \frac{4G_{HM}}{3}} \right)^{-1} - \frac{4G_{HM}}{3} \quad (7)$$

$$G_{dry} = \left(\frac{\frac{\phi}{\phi_c}}{G_{HM} + z_{soft}} + \frac{1-\frac{\phi}{\phi_c}}{G_{min} + z_{soft}} \right)^{-1} - z_{soft} \quad (8)$$

where

$$z_{soft} = \frac{G_{HM}}{6} \left(\frac{9K_{HM} + 8G_{HM}}{K_{HM} + 2G_{HM}} \right) \quad (9)$$

On the other hand, the HS upper bound represents the stiffest possible mineral arrangements, where the grains interact maximally,

leading to the stiffening of the rock (see step-3 in Fig. 5). This behavior is captured by what is referred to as the stiff sand model, which can be expressed as:

$$K_{dry} = \left(\frac{\frac{\phi}{\phi_c}}{K_{HM} + \frac{4}{3}G} + \frac{1-\frac{\phi}{\phi_c}}{K_{min} + \frac{4}{3}G} \right)^{-1} - \frac{4}{3}G \quad (10)$$

$$G_{dry} = \left(\frac{\frac{\phi}{\phi_c}}{G_{HM} + z_{stiff}} + \frac{1-\frac{\phi}{\phi_c}}{G_{min} + z_{stiff}} \right)^{-1} - z_{stiff} \quad (11)$$

where

$$z_{stiff} = \frac{G}{6} \left(\frac{9K + 8G}{K + 2G} \right) \quad (12)$$

The soft-sand and stiff-sand models described above do not account for the additional stiffening of rocks due to cementation. For example, quartz overgrowths, calcite, or dolomite cement at grain contacts or as a coating over the entire grain surfaces strengthen grain bonding and significantly increase the overall stiffness of the rock. Cementation strengthens the grain contacts, effectively “gluing” the grains together and considerably increasing the contact stiffness. In both cases, even a small amount of cement results in a rapid increase in the elastic moduli. To address this, the contact cement model, proposed by Dvorkin et al. (1994), captures the sharp increase in elastic moduli observed in the random packing of spheres when cementation occurs (see step-3 in Fig. 5). The governing equations for this model can be expressed as (Dvorkin et al., 1994; Dvorkin and Nur, 1996):

$$M_c = K_c + \frac{4}{3}G_c \quad (13)$$

$$\alpha = \sqrt{\frac{2}{3} \cdot \frac{\phi_0 - \phi}{1 - \phi_0}} \quad (14)$$

$$\nu_c = \frac{1}{2} \cdot \frac{\frac{K_c}{G_c} - \frac{2}{3}}{\frac{K_c}{G_c} + \frac{1}{3}} \quad (15)$$

$$\nu_s = \frac{1}{2} \cdot \frac{\frac{K_s}{G_s} - \frac{2}{3}}{\frac{K_s}{G_s} + \frac{1}{3}} \quad (16)$$

$$A_N = \frac{2G_c(1-\nu_s)(1-\nu_c)}{\pi G_s(1-2\nu_c)} \quad (17)$$

$$A_T = \frac{G_c}{\pi G_s} \quad (18)$$

$$A_N = -0.024153 \cdot A_N^{-1.3646} \quad (19)$$

$$B_N = 0.20405 \cdot A_N^{-0.89008} \quad (20)$$

$$C_N = 0.00024649 \cdot A_N^{-1.9864} \quad (21)$$

$$A_T = -10^{-2} \cdot (2.26\nu_s^2 + 2.07\nu_s + 2.3) \cdot A_T^{0.079\nu_s^2 + 0.175\nu_s - 1.342} \quad (22)$$

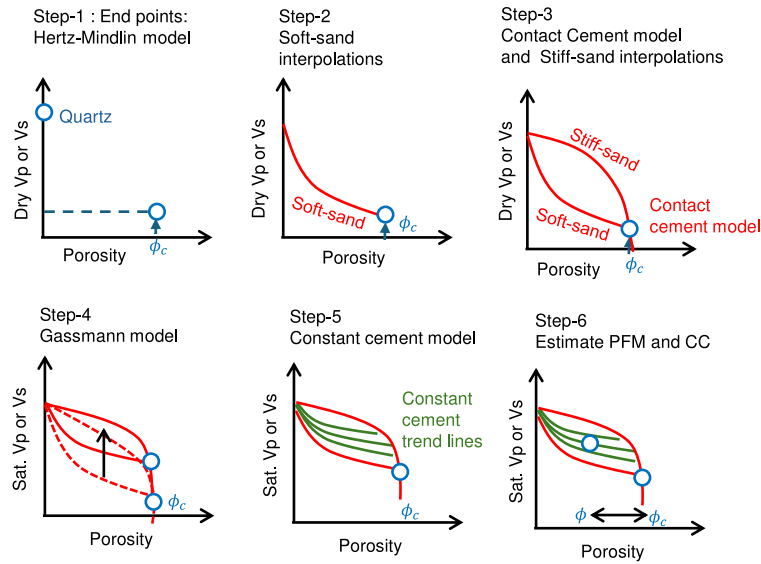


Fig. 5. Rock physics modeling workflow employed in this study, illustrating six key steps. Step 1 defines end points using the Hertz–Mindlin model. Step 2 applies soft-sand interpolations, followed by Step 3, which incorporates the contact cement model and stiff-sand interpolations. Step 4 uses the Gassmann model to account for fluid substitution effects. Step 5 integrates the constant cement model, and Step 6 estimates pore-filling material (PFM) and contact cement (CC).

$$B_T = (0.0573v_s^2 + 0.0937v_s + 0.202) \cdot A_T^{0.0274v_s^2 + 0.0529v_s - 0.8765} \quad (23)$$

$$C_T = 10^{-4} \cdot (9.654v_s^2 + 4.945v_s + 3.1) \cdot A_T^{0.01867v_s^2 + 0.4011v_s - 1.8186} \quad (24)$$

$$S_N = A_N \cdot \alpha^2 + B_N \cdot \alpha + C_N \quad (25)$$

$$S_T = A_T \cdot \alpha^2 + B_T \cdot \alpha + C_T \quad (26)$$

$$K_{\text{dry}} = \frac{n(1 - \phi_0)M_c S_N}{6} \quad (27)$$

$$G_{\text{dry}} = \frac{3K_{\text{dry}}}{5} + \frac{3n(1 - \phi_0)G_c S_T}{20} \quad (28)$$

The contact-cement model presented here is pressure-insensitive. In other words, once the grain aggregate reaches its maximum stiffness due to cementation, further increases in pressure or overburden do not result in additional stiffening (Avseth et al., 2010). However, field observations often contradict this behavior. One possible explanation for this discrepancy is the presence of loose grain contacts or patchy cementation, where not all grains are fully cemented. These localized variations can lead to further stiffening under increased pressure (Avseth and Skjei, 2011; Avseth et al., 2014).

The saturated bulk modulus and shear modulus can be determined using Gassmann's equations (see step-4 in Fig. 5). Gassmann's fluid substitution model is a widely applied method for estimating the bulk modulus of a fluid-saturated rock based on its dry bulk modulus and fluid properties (Gassmann, 1951).

$$K_{\text{sat}} = K_{\text{dry}} + \frac{\left(1 - \frac{K_{\text{dry}}}{K_0}\right)^2}{\frac{\phi}{K_{\text{fl}}} + \frac{(1-\phi)}{K_0} - \frac{K_{\text{dry}}}{K_0^2}} \quad (29)$$

$$G_{\text{sat}} = G_{\text{dry}} \quad (30)$$

Avseth et al. (2000) proposed the constant cement model, which assumes a fixed amount of contact cement at the grain contacts. However, further reductions in porosity are attributed to pore-filling cement or poor sorting. This model is a combination of the contact cement model and the soft sand model. In this framework, the initial reduction in porosity — from the critical porosity downwards — follows the contact cement theory. Beyond this point, the porosity decreases due to the

deposition of cement within the pore spaces, further away from the grain contacts. Practically, these constant cement trend lines can be constructed by virtually increasing the coordination number from the initial value (Mavko et al., 2020b) (see step-5 in Fig. 5).

Moreover, plotting measured data in the velocity-porosity space alongside the contact cement curve and constant cement trend lines allows for quantifying the amounts of both contact cement and pore-filling material (see step-6 in Fig. 5). The procedure follows a sequence of steps, as illustrated in Fig. 6.

1. Identify the endpoint at the critical porosity A and the measured data point C .
2. Identify the constant cement trendline $C - B$ based on the measured data point.
3. Determine point B , where the constant cement trendline intersects with the contact cement model line.
4. Measure the total reduction in porosity from point A to point C , which represents the overall decrease in porosity from the critical porosity to the current porosity. This is a combination of both contact cement and pore-filling material contributions.
5. Calculate the reduction in porosity from point A to point B , which represents the fraction of contact cement (FCC).
6. Calculate the reduction in porosity from point B to point C , which corresponds to the fraction of pore-filling material (FPFM).
7. Finally, the degree of sorting can be calculated as the ratio of observed porosity (porosity at point C) to the end member porosity on constant cement line - porosity at point B .

4. Results and discussion

4.1. Mineralogical distribution

A composite log summarizing all input geophysical logs, along with the mineral and fluid volume predictions, is presented in Fig. 7 for Well-2. To enhance the comparison, the routine core analysis-derived porosity is overlaid on the estimated porosity. Similarly, the XRD-measured weight fractions are converted to volume fractions and overlaid on the mineral volume fraction estimates (see Fig. 7). From this Figure, it is clear that there is a strong agreement between the estimated parameters and the core data, which shows the reliability of the petrophysical evaluation.

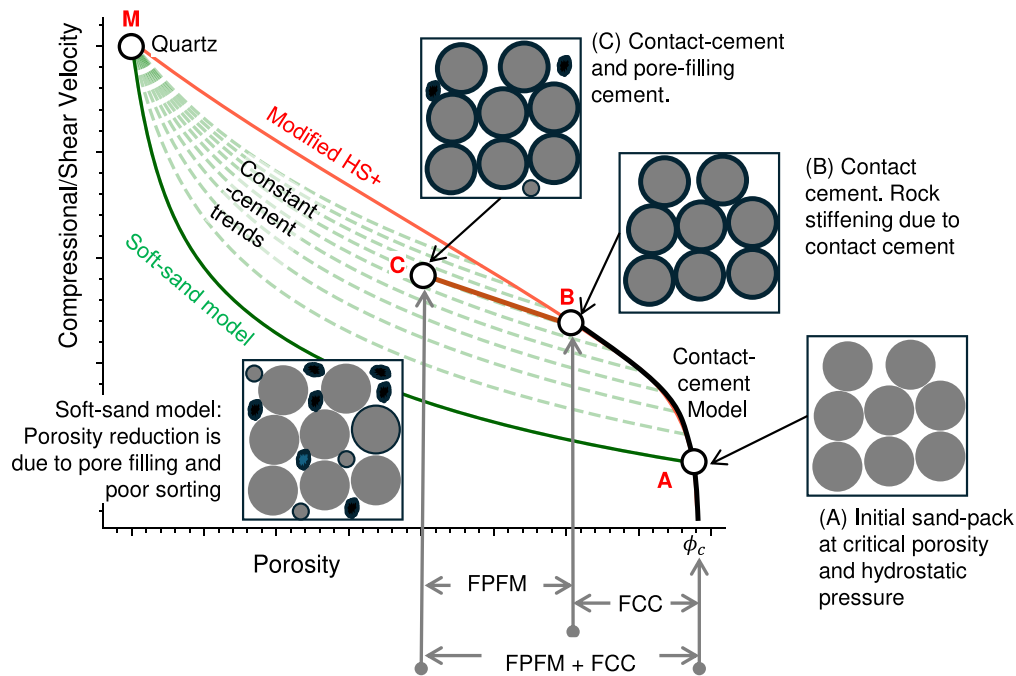


Fig. 6. Schematic representation of the methodology for estimating fraction of pore-filling material (FPFM) and fraction of contact cement (FCC). On the X axis, total porosity and saturated compressional or shear wave velocities are plotted on the Y axis. (M) denotes quartz's velocity at zero porosity (A) Represents the initial sand-pack at critical porosity under hydrostatic pressure. (B) Illustrates rock stiffening due to the addition of contact cement. (C) Depicts the combined effects of contact cement and pore-filling cement. The workflow incorporates the constant cement trend lines, which attribute porosity reduction to pore filling and poor sorting and contact cement line.

Furthermore, the mineral distribution logs for all other wells are provided in the corresponding Fig. 8. From this Figure, it can be interpreted that the Broom Creek Formation is primarily composed of quartz arenite sandstone layers, interbedded with dolomite and thin anhydrite intervals. The sandstone layers exhibit high porosity, up to 0.32, and high permeability, indicative of their excellent reservoir quality. In contrast, the thicker dolomite layers are characterized by very low porosity and are generally impermeable. A minor amount of clay is observed intermittently throughout the interval. Additionally, feldspar is consistently present, along with trace amounts of dolomite and anhydrite within the quartz arenite sandstone layers. These results are consistent with the previous studies conducted in the Broom Creek Formation, which reported similar mineralogical compositions and reservoir characteristics (Pothana et al., 2024; Pothana and Ling, 2025).

However, interpreting the mineralogical distribution alone does not provide sufficient insight into the spatial relationship between these minor minerals and the quartz framework. Specifically, it remains unclear whether these minerals are concentrated at grain contacts or occupy the pore spaces. To address this, we constructed a rock physics modeling template in the velocity-porosity space discussed in the previous section. The mineral volumes estimated during petrophysical evaluation can be used to investigate the correlations between rock's stiffness-porosity evolution during diagenesis. In the next sections, we first present the thin section analysis interpretation; subsequently, we analyze the rock physics modeling results in conjunction with the multi-mineral volumes and the thin section findings.

4.2. Summary of thin section analysis

The thin-section analyses of wells in the Broom Creek Formation provide details into quartz-rich intervals, focusing on grain characteristics, cementation, porosity, diagenetic processes, and depositional

environments. These intervals are characterized by significant variability in textural and mineralogical attributes, reflecting the interplay between depositional and diagenetic processes. Quartz grains in the analyzed sections predominantly range from fine to medium sand size, with occasional coarse grains observed in specific intervals (see Fig. 9). Grain shapes vary from subangular to subrounded, indicating differing degrees of transport and reworking (see Fig. 9). Sorting ranges from moderate to well-sorted, with the best sorting noted in samples interpreted as eolian or high-energy beach sands (see Fig. 10). These textural features suggest deposition under variable energy regimes, with eolian environments producing the most consistent sorting and rounding.

Cementation within quartz-rich intervals is primarily dominated by quartz overgrowths, which play a key role in enhancing the mechanical strength of the rock. This can be observed as some quartz grains appear to have faint, lighter edges surrounded by a thin, dark boundary (dust rim) shown in the plane polarized photomicrographs in Fig. 11. Minor occurrences of dolomite and calcite cement were also noted, particularly in marginal marine environments, where localized diagenetic processes contributed to the partial occlusion of pore spaces. Iron oxide coatings, primarily in the form of hematite, appear as reddish-brown coloration around some grains in the plane-polarized image and as dark rims with bright reflective spots around the quartz grains in the reflected light images shown in Fig. 11. Additionally, the presence of iron oxides imparts a pink to red coloration to the Broom Creek rocks (see Fig. 12). These iron oxide coatings on quartz grains indicate episodes of subaerial exposure and early diagenesis in oxidizing conditions.

The porosity of quartz-rich intervals varies significantly, with effective porosity values generally ranging between 15% and 30%. Intergranular porosity dominates, with secondary porosity being enhanced by dissolution processes as depicted in thin sections as irregular grain boundaries and intergranular voids and, in some cases, the development of microfracture as shown in Figs. 11 and 13. Often, anhydrite pore-filling cement is observed in eolian quartz arenites (see Fig. 15). Quartz-rich zones exhibit relatively high permeability, particularly

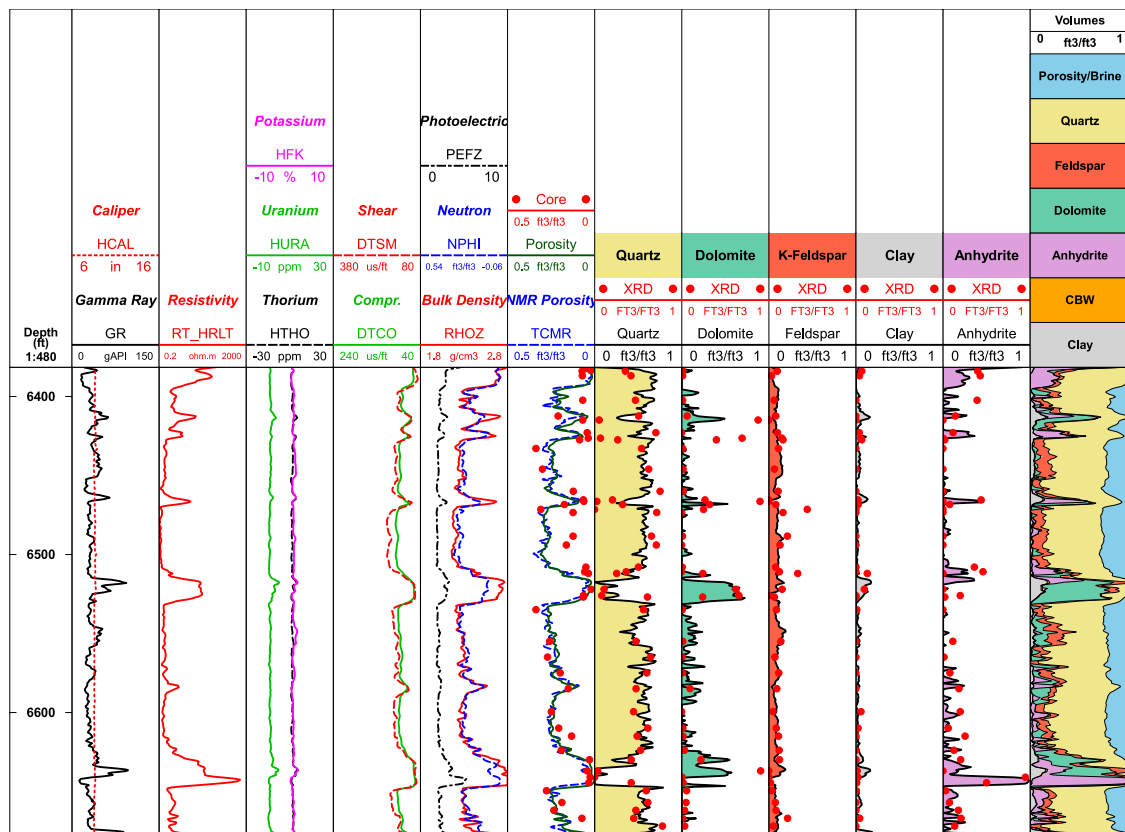


Fig. 7. Multi-mineral petrophysical analysis results for Well-2. This is a vertical well and the depth is measured depth (MD). The logs include gamma ray (GR), caliper (HCAL), resistivity (RT), elemental concentrations (thorium, uranium, and potassium), acoustic properties (compressional and shear travel time: DTCO and DTSM), bulk density (RHOZ), neutron porosity (NPHI), photoelectric factor (PEFZ). Estimated porosity is plotted along with the routine core analysis results and NMR porosity. Estimated mineralogical compositions (quartz, dolomite, feldspar, clay, and anhydrite) are presented along with the XRD-measured volumes.

in intervals with minimal clay content and limited compaction. The preservation of porosity in eolian and beach sandstones highlights their potential as high-quality reservoir intervals. In contrast, porosity in other depositional settings, such as marginal marine environments, is often reduced due to higher compaction and the presence of authigenic clays and anhydrite cement (see Figs. 14 and 15).

Diagenetic processes have markedly influenced the reservoir quality of the Broom Creek Formation. Extensive quartz overgrowths are common, reflecting silica reprecipitation from diagenetic fluids. Evidence of mechanical compaction is observed in the form of elongated tangential grain contacts, and some samples exhibit undulose extinction in quartz grains, indicative of pressure solution as shown in Fig. 16. The variability in diagenetic alteration between wells and intervals emphasizes the heterogeneity of the formation.

The quartz-rich intervals were deposited in a range of environments, including eolian, beach, and marginal marine settings. Eolian deposits are characterized by well-rounded, well-sorted quartz grains, reflecting high-energy conditions that facilitated the removal of finer sediments. These intervals exhibit excellent porosity and permeability, making them particularly significant for reservoir studies (see Fig. 17a). Beach and nearshore environments feature moderately sorted, subangular to subrounded grains, indicative of consistent wave and tidal action (see Fig. 17b). Marginal marine deposits, in contrast, are marked by moderate sorting and subangular grains, with occasional interbedded carbonates and clay-rich zones that reduce reservoir quality (see Fig. 17c). The combination of depositional and diagenetic factors within these environments is integral for understanding the variability in reservoir properties.

4.3. Insights from rock physics modeling

The current study focuses exclusively on the clean sandstone intervals, excluding the tight dolomite and anhydrite layers, for two primary reasons: Firstly, the sandstone intervals exhibit high porosity and permeability, making them the principal storage units within the formation. In contrast, the dolomite and anhydrite layers are characterized by low porosity and are generally impermeable, contributing minimally to reservoir storage capacity. Secondly, the Hertz–Mindlin contact theory is well-suited for modeling the elastic properties of unconsolidated granular materials, such as porous sandstones, where grain-to-grain contacts dominate the mechanical behavior (Mavko et al., 2020b). However, this approach is less appropriate for tight carbonate and anhydrite layers, where the pore structure is more complex, and pores are often isolated within a solid matrix (Regnet et al., 2019). In such cases, effective medium theories, which consider the pores as inclusions within a solid framework, provide a more accurate representation of the elastic properties (Mavko et al., 2020a).

The velocity-porosity rock physics templates are shown in Figs. 18 and 19. Fig. 18 uses color coding to represent the fractions of anhydrite and clay, while Fig. 19 highlights dolomite volume fractions. As discussed earlier, more contact cement makes the rock stiffer but has little effect on porosity and permeability. On the other hand, adding material that fills the pores mainly reduces porosity without changing stiffness much. From the Figures, it is clear that the clean sandstone layers in the Broom Creek Formation have undergone both types of diagenesis—cementation at grain contacts and filling of pore spaces. This dual influence reflects the combined impact of these processes on the burial history of the formation.

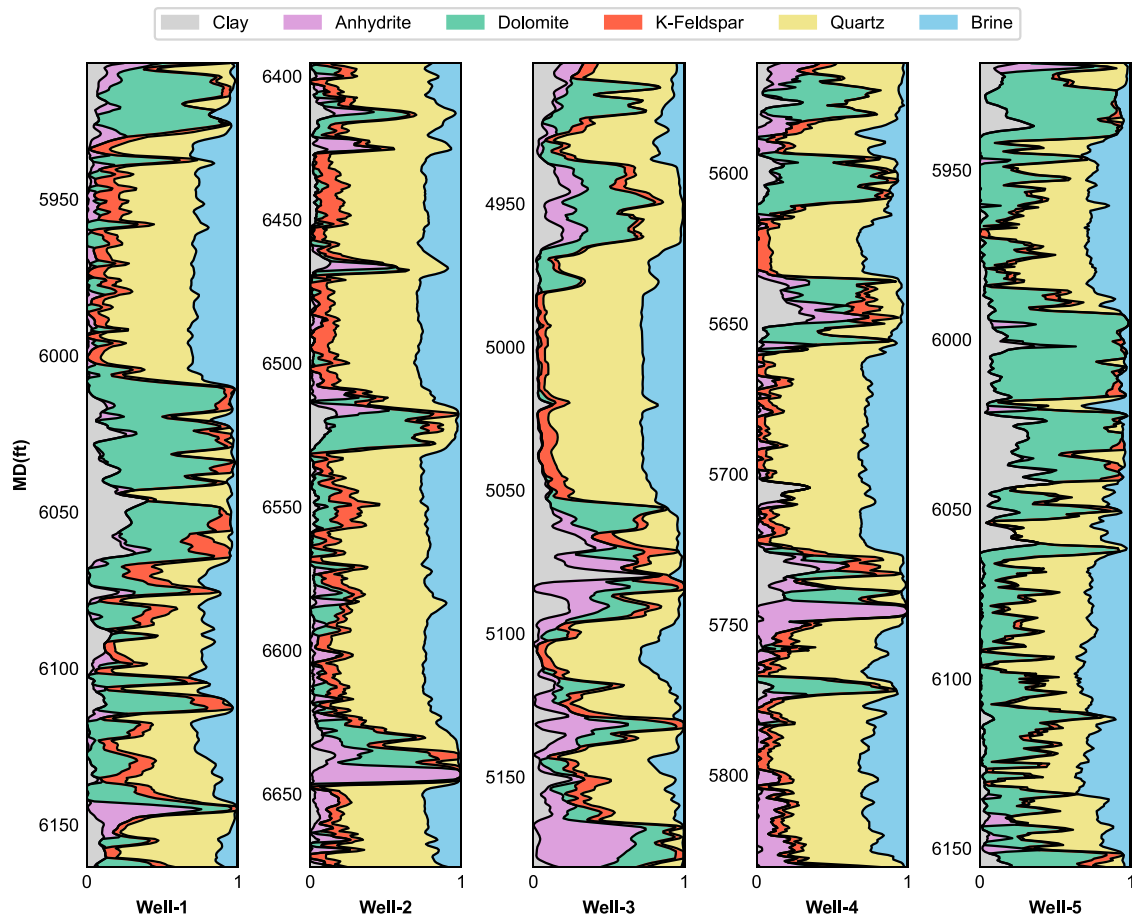


Fig. 8. Multi-mineral petrophysical evaluation results for five vertical wells. The plots illustrate the vertical distribution of mineralogical components, including clay, anhydrite, dolomite, K-feldspar, quartz, and brine, across the measured depth (MD) intervals for each well.

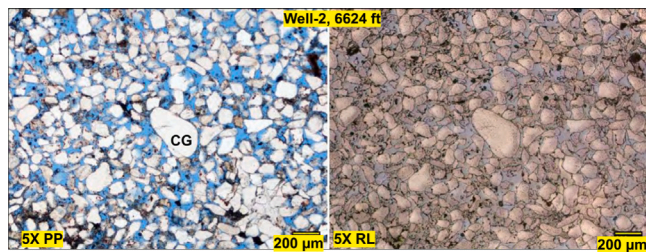


Fig. 9. Photomicrograph analysis showing a 5× plane-polarized light image (left) and a 5× reflected light image (right). The images illustrate fine- to medium-grained quartz with occasional coarse grains, exhibiting predominantly sub-angular to sub-rounded grain shapes.

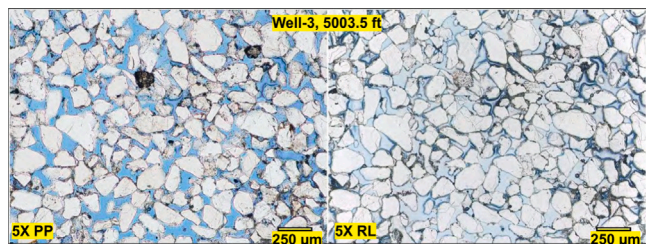


Fig. 10. Photomicrograph analysis showing a 5× plane-polarized light image (left) and a 5× reflected light image (right). The images depict moderately to well-sorted eolian beach sandstone, characterized by quartz grains and an absence of fine-grained material.

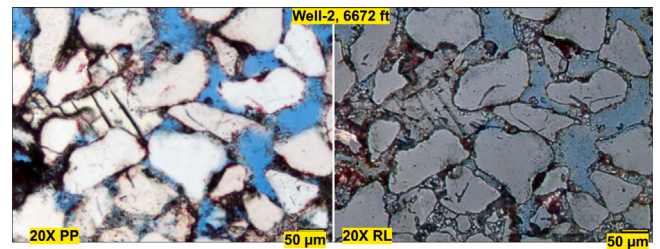


Fig. 11. Photomicrograph analysis showing a 20× plane-polarized light image (left) and a 20× reflected light image (right). The images depict quartz overgrowths, identified by faint, lighter edges surrounding quartz grains with thin, dark boundaries (dust rims) in the plane-polarized light image. Irregular grain boundaries and intergranular voids, as well as, in some cases, the development of microfracture, can also be seen. Iron oxide coatings are visible as reddish-brown coloration around some grains in the plane-polarized image and as dark rims with bright reflective spots around quartz grains in the reflected light image.

Using the methodology described in the previous section, the volume fractions of pore-filling minerals (FPFM) and the fraction of contact cement minerals (FCC) were computed from the cross plots. These parameters were analyzed to investigate the relationships between cementation minerals and their impact on porosity reduction within the sandstone intervals. To better understand which minerals contribute to grain contact cement and which act as pore-filling materials, the FPFM and FCC were plotted against the volume fractions of individual

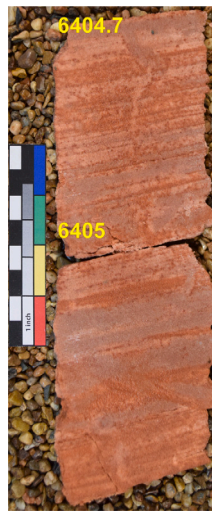


Fig. 12. Core photograph displaying a pink to reddish coloration attributed to iron oxide coatings, predominantly in the form of hematite.

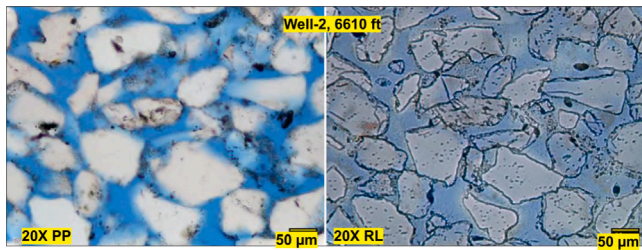


Fig. 13. Photomicrograph analysis showing a 20x plane-polarized light image (left) and a 20x reflected light image (right). The images reveal dominant intergranular porosity, with secondary porosity enhanced by dissolution processes, evidenced by irregular grain boundaries and intergranular voids visible in the thin sections.

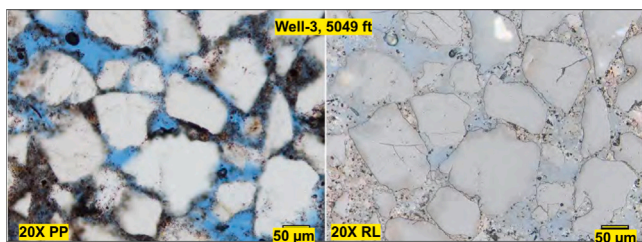


Fig. 14. Photomicrograph analysis showing a 20x plane-polarized light image (left) and a 20x reflected light image (right). The images highlight the presence of authigenic clays and carbonate cement occupying the intergranular pore spaces.

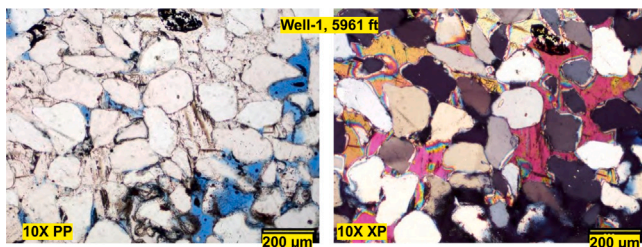


Fig. 15. Photomicrograph analysis showing a 10x plane-polarized light image (left) and a 10x cross-polarized light image (right). The images highlight the presence of anhydrite (High interference color) filling the intergranular pore space among the quartz grains.

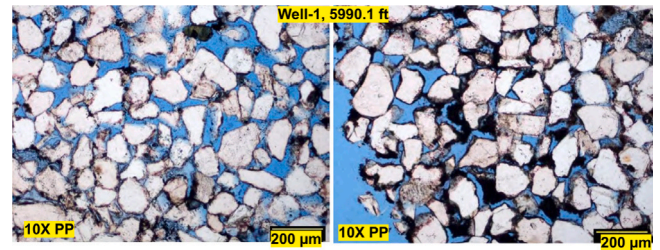


Fig. 16. Photomicrograph analysis showing a 10x plane-polarized light image (left) and a 10x reflected light image (right). The images display elongated tangential grain contacts and undulose extinction in some quartz grains, indicative of pressure solution processes.

minerals, as shown in Figs. 20 and 21. Each scatter plot includes a best-fit line and correlation coefficient value (r); the correlation coefficients were calculated using Pearson's correlation (see Eq. (A.1)).

From the photomicrograph analysis, it is evident that quartz and feldspar are the grain-forming minerals in the reservoir layers, while clay, anhydrite, and dolomite are the cementing materials in addition to quartz overgrowth. The results indicate that clay and anhydrite volumes exhibit a consistent positive correlation with the FPFM across the five wells, as seen in Fig. 20, suggesting that these minerals play a significant role in reducing porosity by occupying pore spaces. In contrast, dolomite and feldspar volumes show no significant correlation with the FPFM, indicating a limited impact on pore filling by these minerals. As shown in Fig. 21, dolomite displays a weak positive correlation with the FCC consistently among all five wells, suggesting its contribution to contact cement formation. However, no meaningful correlations were observed between FCC and other cementing minerals. These findings emphasize the different roles of various minerals in controlling porosity and cementation within the studied sandstone intervals.

We utilized the observations from these correlation plots to further investigate the depth-wise relationships between cementation materials and mineral compositions. Specifically, we plotted the FPFM and the degree of sorting across the Broom Creek intervals alongside the combined volumes of anhydrite and clay, as shown in tracks (d) and (f) of Figs. 22, B.25 B.26, B.27, and B.28. These plots reveal a strong correlation between the volume of anhydrite and clay and the FPFM. This observation suggests that anhydrite and clay primarily function as pore-filling minerals within the sandstone rather than forming contact cement around the quartz grains. This interpretation is also supported by the thin section analysis results, where it is observed that the presence of anhydrite and clay in the intergranular pore spaces (see Figs. 14 and 15). The anhydrite pore-filling could be caused by sulfate-rich brines moving into aeolian sandstone sediments, often deposited in arid environments with evaporative conditions. During diagenesis, anhydrite precipitates from this brine, initially as gypsum, which dehydrates to anhydrite under burial conditions exceeding 50 °C (Hardie, 1978). A similar observation was reported in Rhaetian sandstones of aeolian origin, where anhydrite was found filling intergranular pore spaces (Baermann et al., 2000).

The contact cement in the Broom Creek Formation can be interpreted as quartz overgrowth in the aeolian sandstone layers and quartz overgrowth and dolomite precipitation at grain contacts in the nearshore marine sandstone layers. A limitation of multi-mineral petrophysical analysis is its inability to distinguish between detrital quartz grains and quartz overgrowths, as it only provides the total quartz volume fraction. Identifying quartz overgrowths requires photomicrograph or SEM image analysis. The stiffening of the aeolian sandstone layers, supported by the absence of significant additional minerals and confirmed by thin section analysis showing quartz overgrowth (for example, see Fig. 11), suggests that quartz overgrowth is the primary

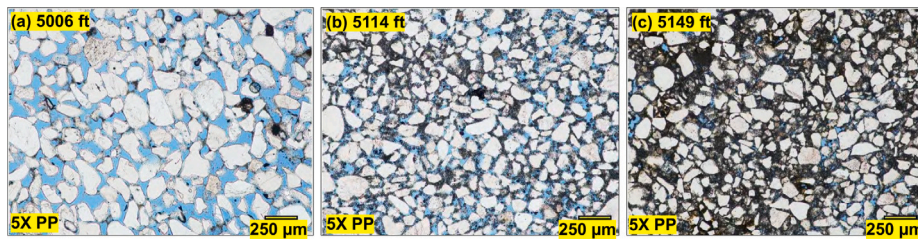


Fig. 17. Photomicrograph analysis at 5× plane-polarized light showing quartz-rich intervals from eolian, beach, and marginal marine settings. Eolian deposits (left) exhibit an absence of fines, well-rounded, well-sorted grains with high porosity and permeability. Beach and nearshore deposits (middle) show moderately sorted, subangular to subrounded grains, while marginal marine deposits (right) feature moderate sorting, subangular grains, and occasional carbonate or clay-rich zones.

contributor to the enhanced stiffness. This highlights the critical need to combine multiple methods to accurately delineate spatial mineralogical distributions and understand their impact on reservoir properties.

The sorting indicator (SI) is defined as the ratio of the observed porosity to the end-member porosity on the constant-cement line (Avseth et al., 2010):

$$SI = \frac{\phi}{\phi_{\text{intersect}}} \quad (31)$$

Here, ϕ is the total observed porosity at point C in Fig. 6, and $\phi_{\text{intersect}}$ is the porosity at the end member on the constant-cement line at point B. By definition, SI increases when $\phi_{\text{intersect}}$ decreases, and vice versa. Because the porosity reduction from B to C reflects both poor grain sorting and pore-filling cement, SI must be used with caution—especially in intervals containing non-contact cements such as anhydrite and clay. Figs. 22, B.25, B.26, B.27 and B.28 plot SI alongside volumes of non-contact cements for five wells. In the clean sandstone intervals of the upper Broom Creek Formation, SI values approach 1, indicating well-sorted grains. In the lower sections, SI decreases, reflecting poorer sorting compounded by higher proportions of clay and anhydrite, which further lower the indicator.

These variations in sorting are closely tied to the depositional environments of the Broom Creek Formation. Aeolian sand deposits, typically found in dune fields and sand sheets, are well-sorted due to the selective transport mechanism of wind. Wind effectively mobilizes and transports particles of similar size, predominantly fine to medium sand grains, while larger or denser particles are excluded due to their higher settling velocities. This process produces texturally mature and homogeneously graded deposits (Boggs, 2012a). Nearshore sandstones, on the other hand, display moderate to well-developed sorting but with greater variability compared to aeolian deposits. These environments are shaped by dynamic processes such as wave action, tides, and currents, which selectively transport and deposit sediments. While waves and currents rework sediments and improve sorting, variations in sediment supply and biogenic mixing introduce heterogeneities (Boggs, 2012a,b). These rock physics modeling results are also consistent with the photomicrograph analysis results, which are shown in Figs. 10 and 17.

4.4. Uncertainty analysis

Uncertainty analysis was conducted using a Monte Carlo approach to assess the sensitivity of input end points on the pore filling cement (FPFM) and the contact cement (FCC) volume fractions. A uniform random sampling method was employed to generate realizations of the input parameters, which included critical porosity, quartz bulk modulus, and quartz shear modulus. These parameters were varied within $\pm 5\%$ of their reference values to capture the range of possible outcomes under input uncertainty. These parameters directly influence the effective moduli computed and ultimately the FCC and FPFM estimates. A total of 10,000 iterations were run to capture the range of possible outcomes under input uncertainty. The analysis provided

distributions of porosity (FCC and FPFM) and sorting metrics, from which statistical descriptors such as mean, standard deviation, and 5th–95th percentiles were extracted. Fig. 23 illustrates the distributions of the input parameters (critical porosity, quartz bulk modulus, and quartz shear modulus) and the output parameters (contact cement, pore filling cement, and sorting). The histograms show the range and frequency of each parameter's values, highlighting the variability introduced by the Monte Carlo simulations. The combined uncertainty analysis indicates that the mean porosity for cemented components (FCC) is approximately 5.3% with a standard deviation of 1.0%, ranging between 0.5% and 14.9%. In contrast, non-cemented components (FPFM) exhibit a higher mean porosity of 7.9% and a wider spread (standard deviation of 1.9%), with values spanning from 0.2% to 13.7%. Sorting values cluster more tightly, with a mean of 0.776 (dimensionless), a standard deviation of 0.044, and a range from 0.66 to 0.99. The 5th to 95th percentile intervals for FCC, FPFM, and sorting are [0.041, 0.069], [0.047, 0.107], and [0.714, 0.850], respectively, capturing the core spread of the modeled uncertainty.

The sensitivity coefficients (r) were calculated using Spearman's rank correlation method, which is robust against non-normal distributions and outliers. The results, shown in Fig. 24, indicate that FCC is most sensitive to critical porosity ($r = 0.71$), with a moderate effect from quartz shear modulus ($r = 0.40$) and a smaller effect from quartz bulk modulus ($r = 0.25$). FPFM is dominated by critical porosity ($r = 0.93$); sensitivity to the quartz elastic end points is weak (≥ 0.16). Sorting mirrors FPFM; it is driven by critical porosity ($r = 0.93$) and only weakly influenced by quartz elastic moduli.

The correlation between FCC/FPFM and mineral volumes may be affected by uncertainty in petrophysical inversion, particularly from log noise and XRD calibration mismatch. Log-based mineral volume estimates can vary due to borehole effects, tool resolution, or interpretation biases, while XRD-derived fractions are sensitive to grain size, sample heterogeneity, and phase identification limits (Doveton, 1994). These uncertainties can weaken the correlation strength between computed cement fractions and observed mineral phases.

4.5. Practical implications for CO₂ sequestration

Injected CO₂ dissolves into brine, forms carbonic acid, and drives dissolution–precipitation reactions that modify porosity, permeability, and frame stiffness; these changes control injectivity, fracture risk, and the reservoir's seismic response (Kaszuba et al., 2003; Palan-dri and Kharaka, 2004; Rutqvist, 2012). In Broom Creek sandstones, the dominant cements and minor phases identified in this study — quartz overgrowths, dolomite, anhydrite, feldspar, and clays — govern how the pore system and mechanical framework evolve under CO₂ exposure.

Carbonic-acid brines increase quartz dissolution rates modestly, especially at higher temperature and low pH; effects are slow relative to carbonates, so direct frame weakening by quartz dissolution is limited on operational time scales, although minor porosity increases may occur (Zhao et al., 2015; Knauss and Wolery, 1988; Lasaga, 1984).

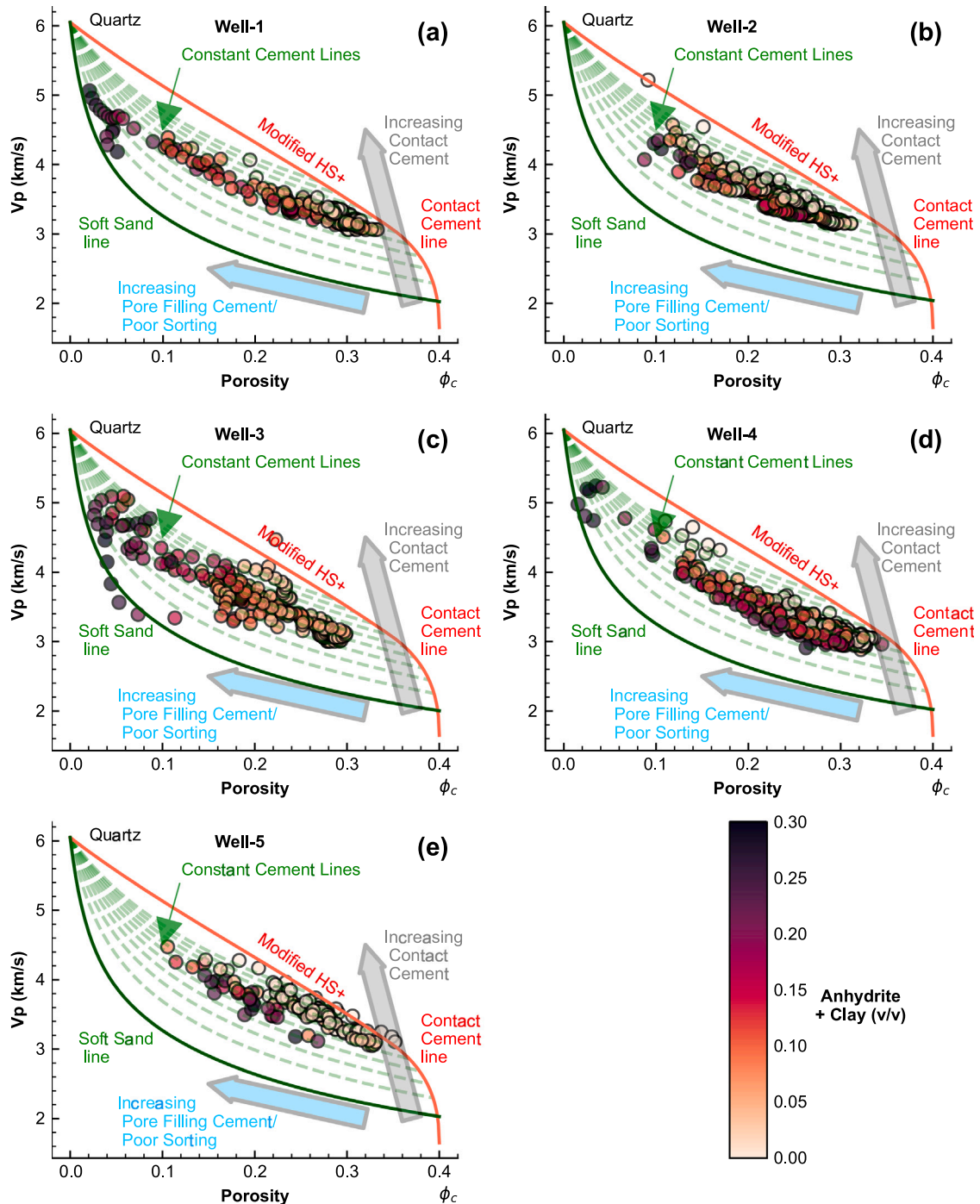


Fig. 18. Saturated compressional wave velocity data plotted against total porosity with superimposed diagnostic rock-physics models, including constant cement trend lines and the contact cement trend line. These models are used to quantify the fractions of pore-filling material and contact cement. Each subplot represents data from an individual well, with data points color-coded based on the volume of anhydrite and clay (as shown in the color bar) to illustrate their relationship with diagenetic trends.

Dolomite dissolves readily in CO_2 -acidified brines and the rate accelerates with lower pH and higher temperature; loss of dolomite at grain contacts could potentially reduce local stiffness and cohesion, with permeability increases possible where dissolution channels form, but fines mobilization and localized strain may also raise fracture susceptibility (Lamy-Chappuis et al., 2016; Berrezueta et al., 2023; Lasaga, 1984). Anhydrite is less reactive than carbonates under many CO_2 -brine conditions, yet gypsum-anhydrite transitions are thermally controlled and

sulfate can re-precipitate as scale when Ca^{2+} liberated by carbonate dissolution mixes with sulfate-bearing brine, potentially clogging pore throats and creating flow barriers (Isah et al., 2022; Hangx et al., 2009). Feldspar shows preferential dissolution in CO_2 -acidified brines, with Ca-rich plagioclase dissolving fastest; long-duration in-situ P-T experiments on Ketzin sandstones document corrosion of intermediate plagioclase and K-feldspar with concurrent precipitation/stabilization of albite, alongside anhydrite dissolution (Fischer et al., 2010).

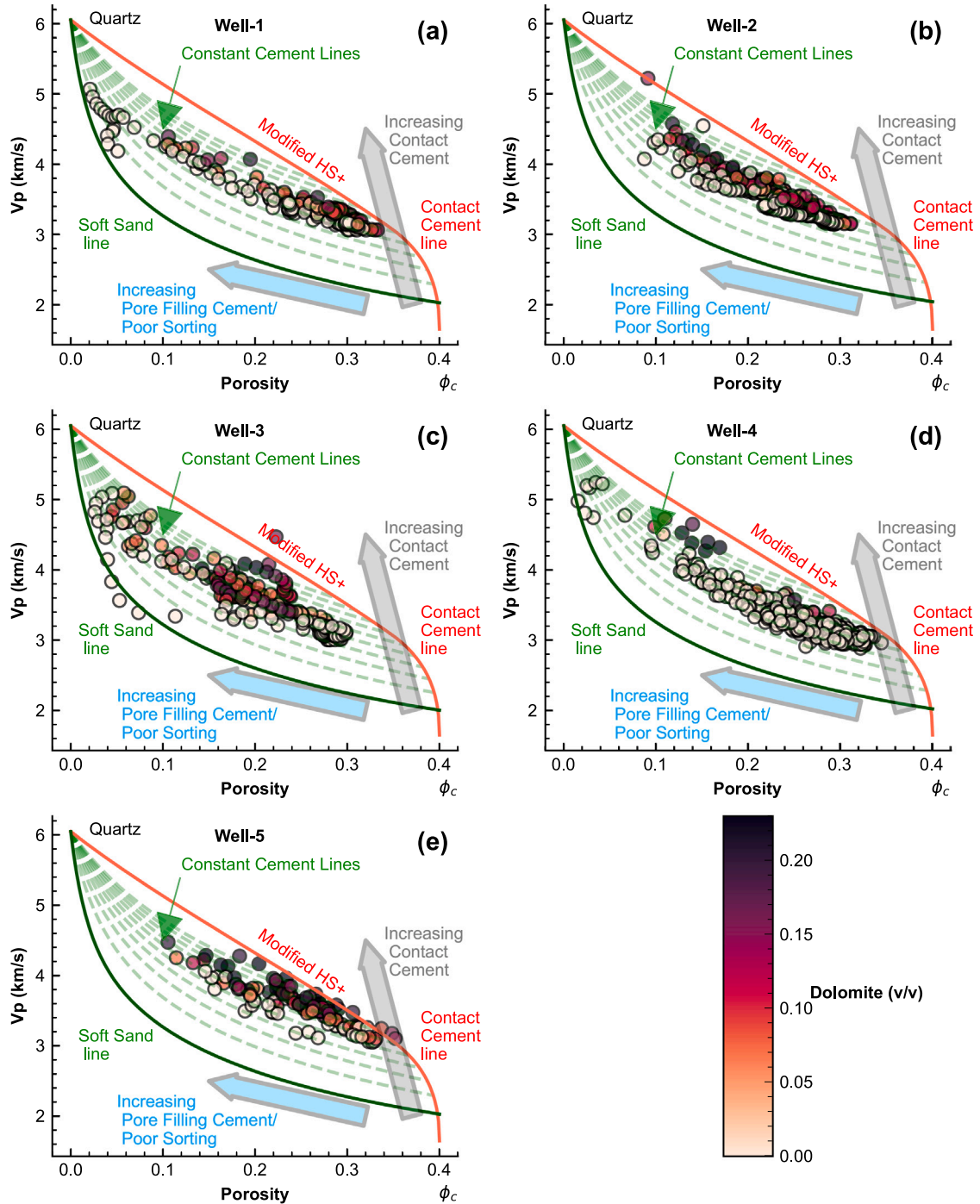


Fig. 19. Saturated compressional wave velocity data plotted against total porosity with superimposed diagnostic rock-physics models, including constant cement trend lines and the contact cement trend line. These models are used to quantify the fractions of pore-filling material and contact cement. Each subplot represents data from an individual well, with data points color-coded based on the volume of dolomite (as shown in the color bar) to illustrate their relationship with diagenetic trends.

Core-flooding further indicates that with increasing temperature and CO_2 the susceptibility sequence is carbonates > feldspar >> quartz; secondary phases (calcite, ankerite, hematite, albite) precipitate and clay assemblages evolve, reorganizing throat size distributions, lowering permeability, and increasing displacement pressure (Zhao et al., 2015). Clay- CO_2 -brine interactions are system-dependent: clay-rich shales exposed to CO_2 -brine undergo dissolution of illite/chlorite and formation of smectite, with associated carbonate precipitation (Alemu

et al., 2011). Dissolution dominates in phlogopite with secondary amorphous silica and kaolinite precipitates that alter wettability and block pores (Shao et al., 2010); kaolinite and montmorillonite subjected to supercritical CO_2 show aggregation, capillary-driven contraction, and desiccation cracks, decreasing effective porosity and increasing breakthrough risk (Espinoza and Santamarina, 2012).

For injectivity and near-wellbore behavior, carbonate-cement dissolution can transiently increase permeability and injectivity, whereas

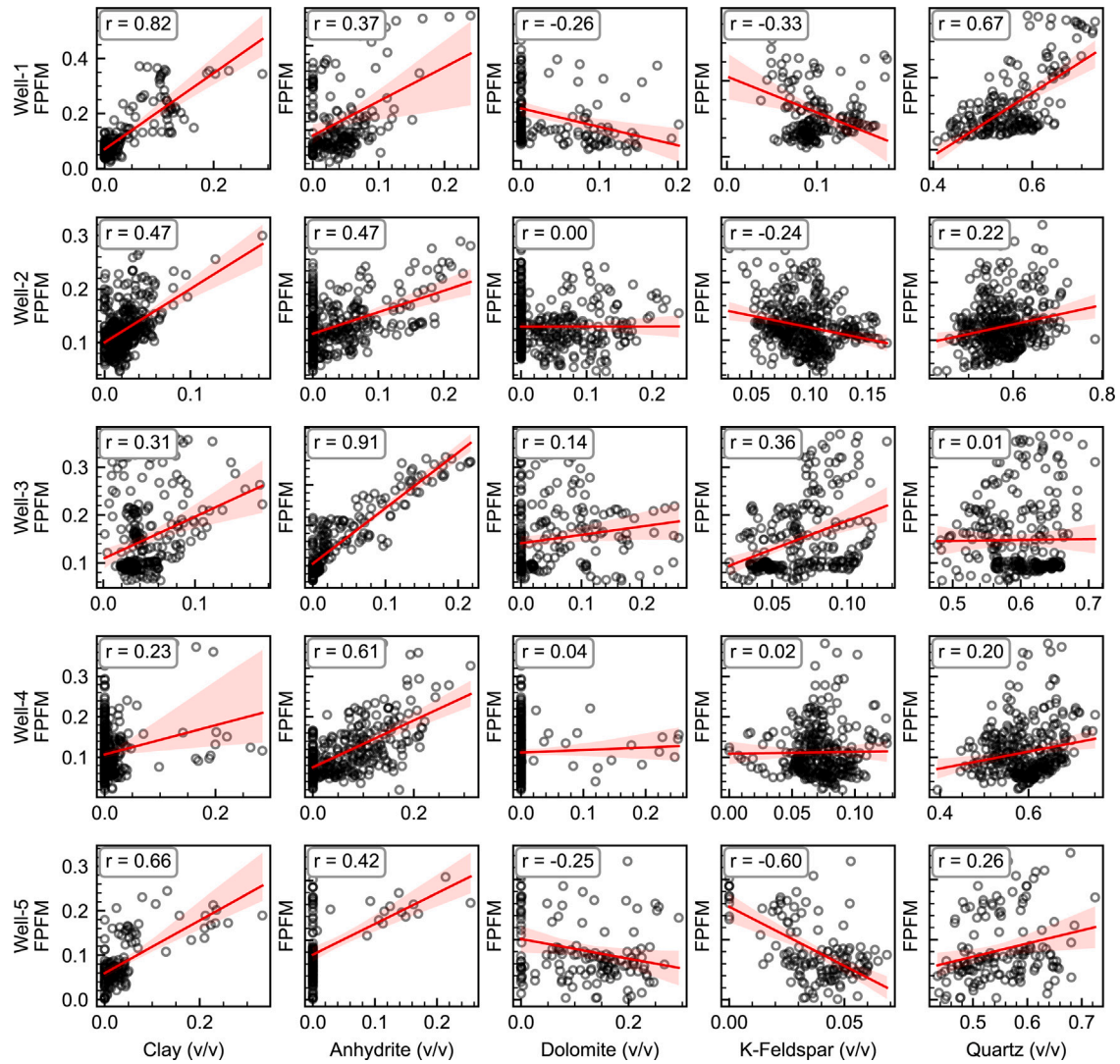


Fig. 20. Exploratory data analysis of the relationship between the fraction of pore-filling material (FPFM) and individual mineral volumes. Each subplot presents a linear regression fit (red line) and its 99% confidence interval (shaded band). The Pearson correlation coefficient r is annotated in the upper left of each panel. Rows correspond to data from individual wells. The analysis shows a consistent positive correlation between FPFM and the volumes of clay and anhydrite cement.

secondary precipitation of carbonate, sulfate, or clay can reverse those increments by occluding pore throats (Seyyedi et al., 2020; Singurindy and Berkowitz, 2003). An additional risk is salt precipitation during dry-out when injecting dry or drier-than-formation CO_2 : halite can precipitate rapidly near the wellbore and severely reduce permeability; field and pilot observations support this mechanism, and mitigation relies on managing water with the injected stream and monitoring for salt with appropriate surveillance (Bacci et al., 2013; Pruess and Müller, 2009). For mechanical integrity, dissolution of dolomite or calcite contact cements can reduce frame stiffness and failure strength locally; effects are facies-dependent and likely greatest in carbonate-rich streaks, whereas cement-poor eolian quartz arenites are less sensitive. Pressure remains the dominant control on fault reactivation and induced seismicity, but any geochemical softening narrows strength margins; hence, rate and pressure management should be designed accordingly, and the pore filling and grain contact cement estimates would help in this case (Luo et al., 2022a). For seismic monitoring, geochemical alteration changes velocities through both fluid substitution and frame modification. Dissolution of stiff carbonate cements tends to decrease V_p and V_s at fixed saturation, whereas precipitation of stiff cements has the opposite effect (Adam et al., 2015). Such trends have

been resolved with 4D seismic in CO_2 storage settings, and the velocity-porosity templates developed here provide useful priors for time-lapse feasibility studies and for discriminating saturation/pressure effects from geochemical stiffening or softening in inversion (Kumar et al., 2008; Ma et al., 2016). Incorporating mineralogical maps and pore filling and grain contact cement estimates can improve the interpretability of time-lapse signatures in Broom Creek formation.

4.6. Limitations and future work

- Quartz overgrowth was inferred from visual petrography. Separation of overgrowths from detrital quartz cannot be achieved with volumetric well logs. Quantitative imaging using SEM with segmentation or point counting is required to estimate the overgrowth fraction with uncertainty. The measured fraction should be compared with plug ultrasonic and static moduli to evaluate the contribution to stiffness.
- Quantitative grain-size and sorting statistics are not available. Automated thin-section image analysis with scale calibration will enable computation of grain-size distributions and sorting coefficients, such as Folk-Ward metrics. These statistics will reveal

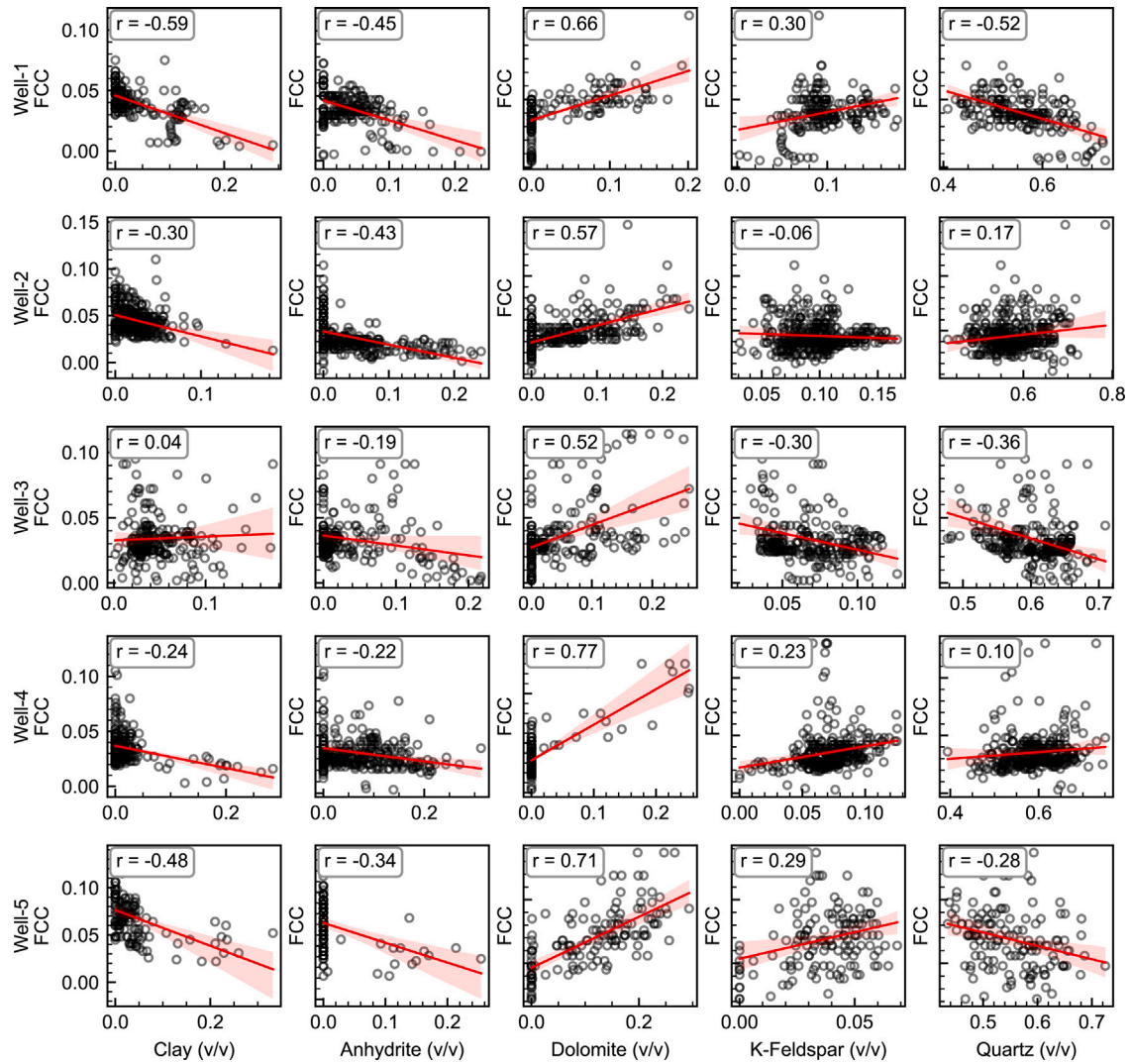


Fig. 21. Exploratory data analysis of the relationship between the fraction of contact cement (FCC) and individual mineral volumes. Each subplot presents a linear regression fit (red line) and its 99% confidence interval (shaded band). The Pearson correlation coefficient r is annotated in the upper left of each panel. Rows correspond to data from individual wells. The analysis shows a consistent positive correlation between FCC and the volumes of dolomite.

relationships to facies, sorting indicator or pore filling material and core permeability.

- The mixing regimes that drive sulfate, carbonate, or clay precipitation are unknown, and the near-wellbore versus far-field localization is uncertain. Tracer-supported pilots need to be conducted. Micro-CT and SEM will map when and where scaling occurs and quantify its impact on flow.
- The cement-loss level that measurably reduces stiffness and increases fracture risk in cement-rich facies is not quantified. Tri-axial tests conducted before and after CO₂-brine exposure on representative facies will provide the needed constraints. The results should be compared with contact-cement maps to identify weakening thresholds.
- The scope is limited to clean sandstone, and transfer to tight carbonate and anhydrite intervals is uncertain. Modeling needs to be extended to those rocks using effective-medium theories for isolated or patchy porosity. These extensions should be validated against core measurements or other observations.
- A single-phase, brine-saturated model was used. This choice provides a consistent baseline but may not represent two-phase

conditions during injection. Partial CO₂ saturation, patchy saturation, and capillary effects could change velocities, impedances, and the interpretation of porosity and stiffness. Future work could explore fluid-substitution cases and compare with lab corefloods and time-lapse logging or VSP to bound the impact.

- Dynamic validation of predicted CO₂ effects is missing. CO₂-brine corefloods at reservoir pressure and temperature should be performed with pre- and post- μ CT, QEMSCAN, and ultrasonics. Pilot-scale 4D monitoring will test the template predictions.
- Rate and pressure management strategies are not yet evaluated within coupled physics. These strategies should be evaluated in coupled flow- geochemistry- geomechanics models. Links between possible geochemical softening and safe operating envelopes must be established for candidate injection scenarios.

5. Conclusions

This study evaluates the Broom Creek Formation, an important candidate for CO₂ sequestration in the Williston Basin. By integrating multi-mineral petrophysical analysis, thin section evaluation, and rock

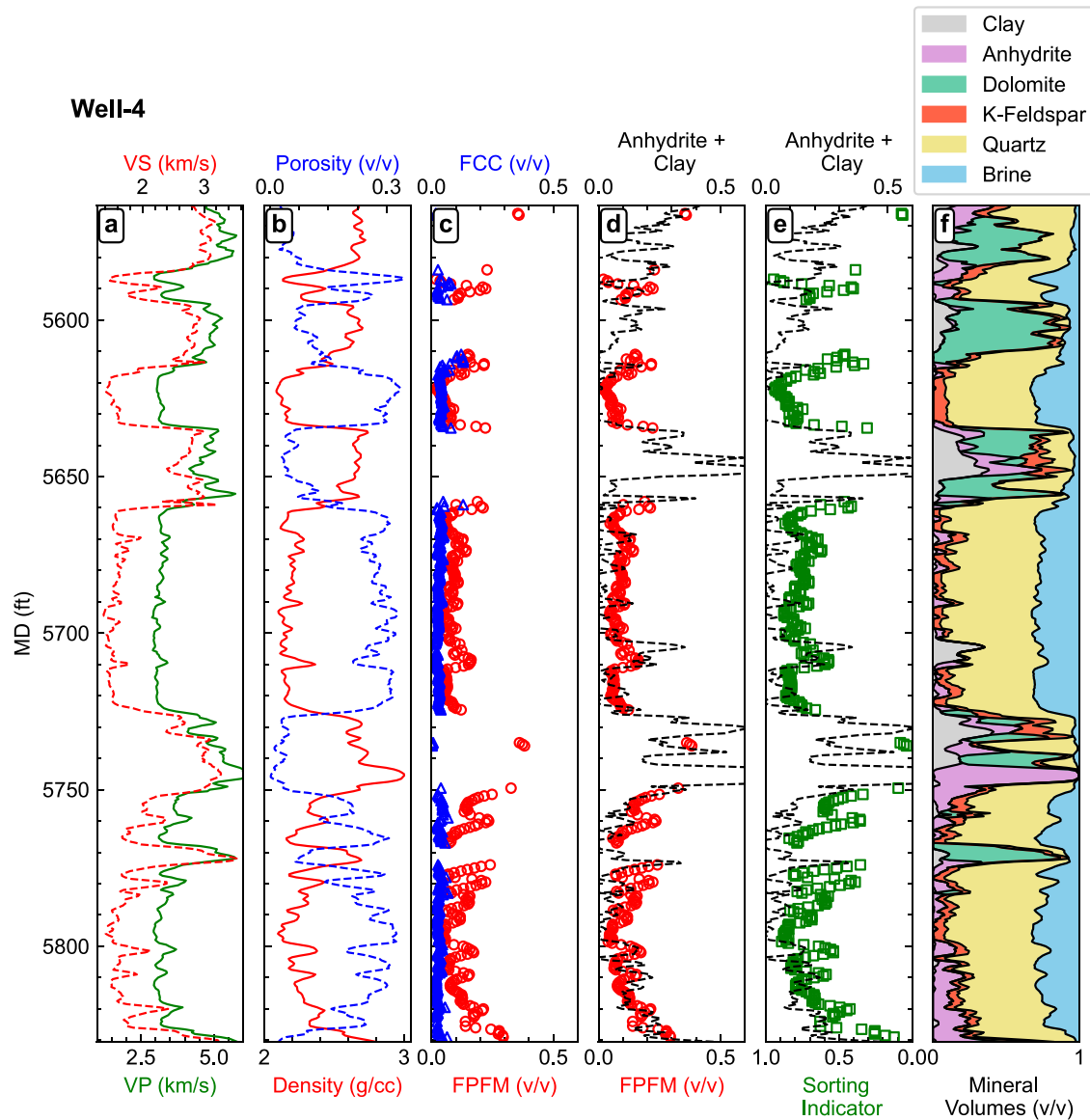


Fig. 22. Composite log, rock physics modeling, and petrophysical analysis results for Well-4. Track (a) displays compressional and shear wave velocities. Track (b) presents porosity and bulk density. Track (c) illustrates the fractions of pore-filling material (FPFM) and contact cement volume (FCC). In track (d), FPFM is shown as scatter points overlaid on the combined volume fractions of anhydrite and clay. Track (e) depicts the degree of sorting (as defined in Eq. (31)) as scatter points, with the combined anhydrite and clay plotted as a continuous line. Track (f) showcases the multi-mineral distribution, with each mineral represented by a distinct color as indicated in the legend.

physics modeling, we have delineated the influence of diagenetic processes on the formation's mechanical and petrophysical properties. The results emphasize the dual role of diagenetic cements: contact cements, such as quartz overgrowths and dolomite, enhance the rock stiffness and structural integrity, while pore-filling materials, including anhydrite and clay, reduce porosity and permeability. These results indicate the interplay between depositional environments and diagenesis, which collectively shape reservoir quality.

Quartz and feldspar constitute the primary grain framework, with quartz overgrowths and dolomite serving as the main grain-contact cement, while anhydrite and clay function as pore-filling materials. Quartz-rich intervals deposited in high-energy aeolian and beach environments exhibit excellent reservoir quality, whereas intervals with extensive pore-filling material in marginal marine settings display reduced porosity and injectivity.

The study shows the importance of integrating geological, petrophysical, and rock physics models to better understand the evolution of

elastic properties and diagenetic trends. The velocity-porosity modeling approach, combined with mineralogical data, provided information about the spatial distribution of cements and their contributions to reservoir stiffness. The ability to differentiate between contact and non-contact cement is essential for predicting the reservoir's response to CO₂ injection and ensuring long-term storage stability.

These findings offer strategic implications for the success of CO₂ sequestration projects, particularly in deep saline aquifers like the Broom Creek Formation. The information gained here not only enhances our understanding of reservoir behavior under varying diagenetic conditions but also provide a methodological framework applicable to analogous geological formations. Continued integration of thin section analysis, modeling, and field-scale data is recommended to refine predictions of reservoir performance and ensure the sustainability of CO₂ storage operations.

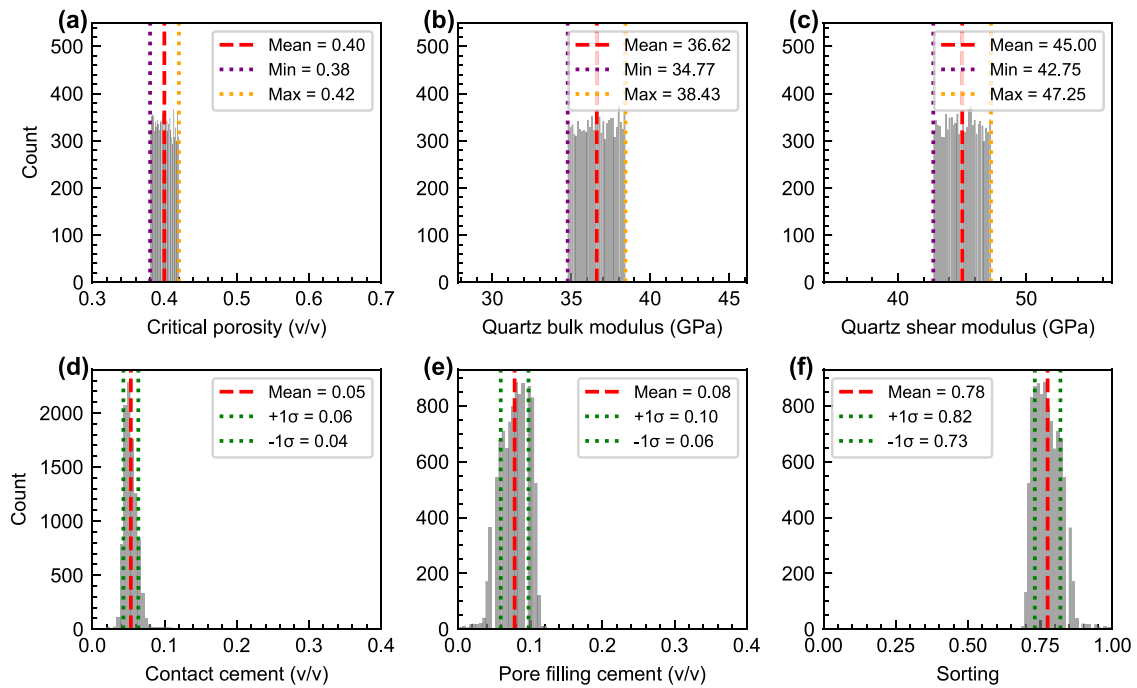


Fig. 23. Distributions of input and output parameters for Well-2. Top row shows histograms of critical porosity, quartz bulk modulus, and quartz shear modulus; bottom row shows histograms of contact cement, pore filling cement, and sorting. Mean and standard deviation are indicated for each parameter.

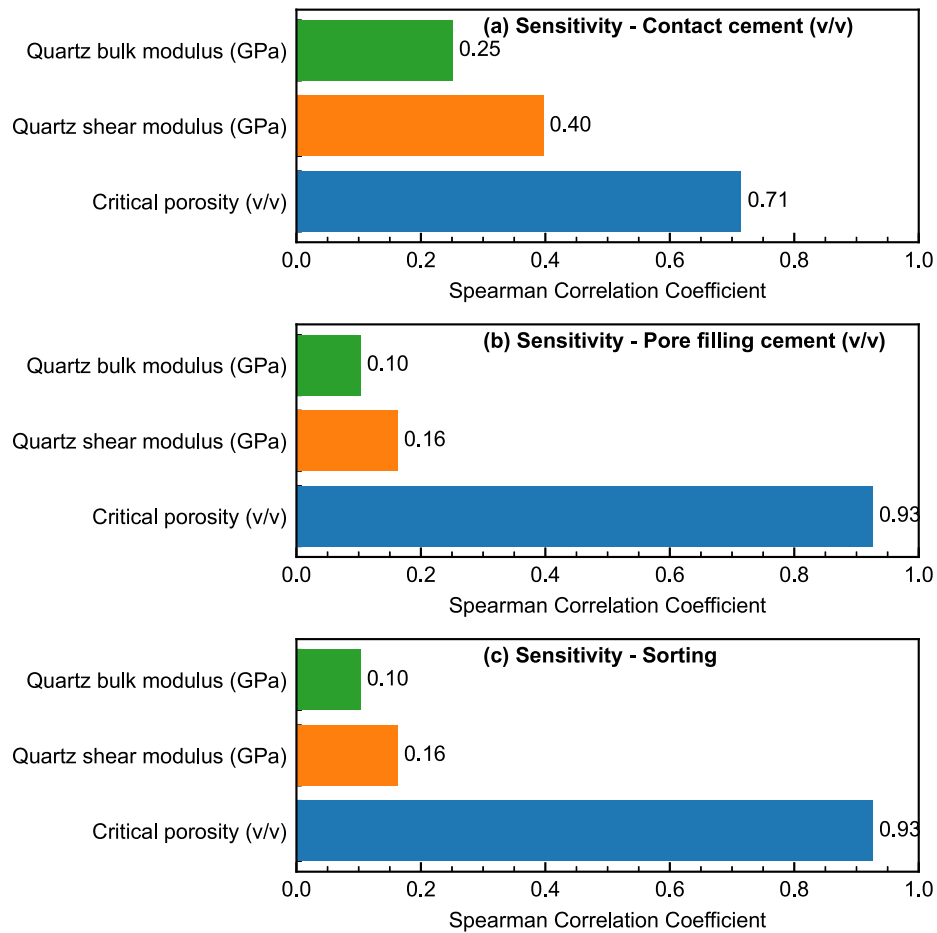


Fig. 24. Sensitivity of output parameters (contact cement, pore filling cement, and sorting) to input variables, based on Spearman correlation coefficients. Bars represent the relative influence of each input on the outputs.

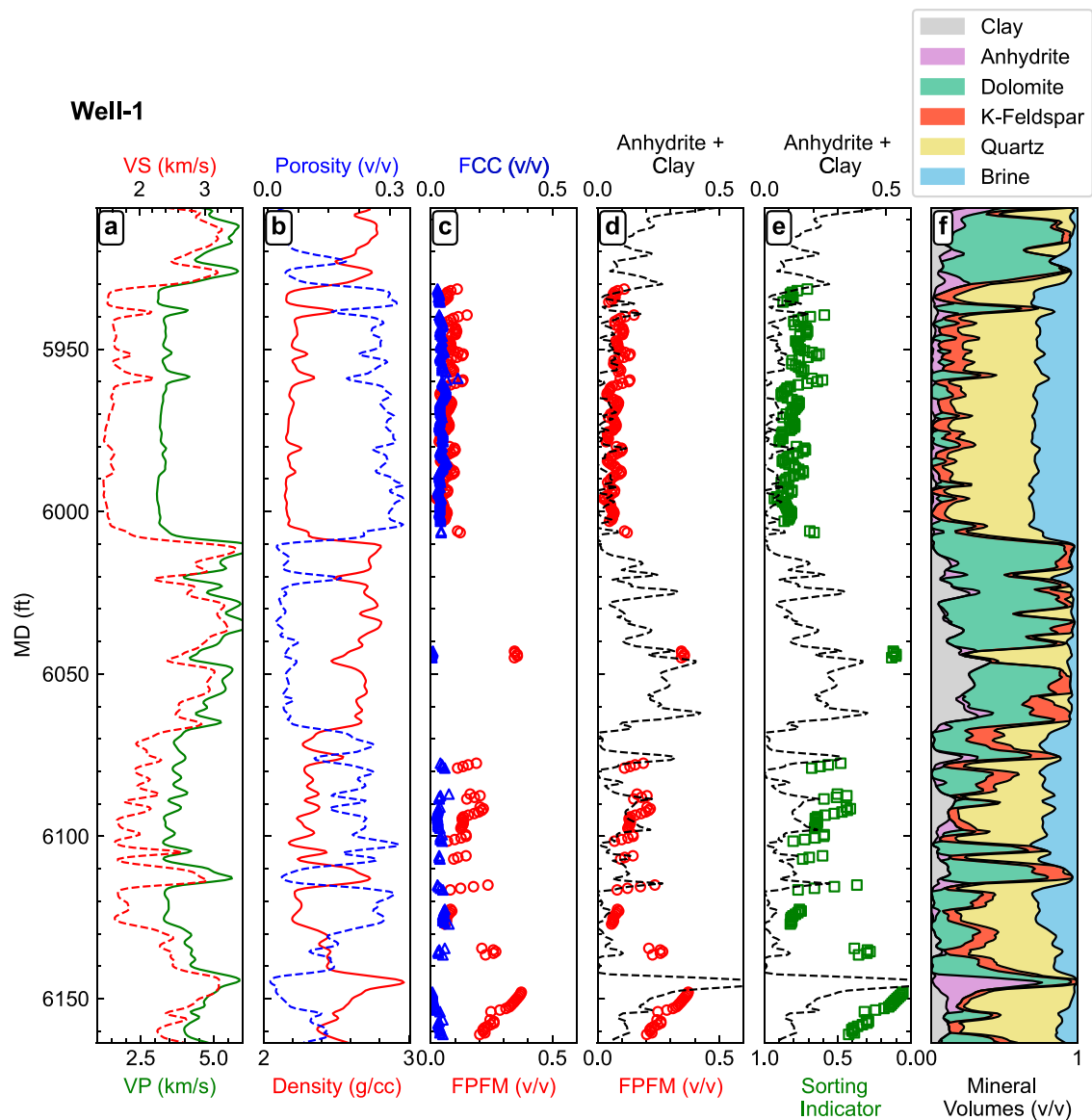


Fig. B.25. Composite log, rock physics modeling, and petrophysical analysis results for Well-1. Track (a) displays compressional and shear wave velocities (V_p and V_s). Track (b) presents porosity and bulk density. Track (c) illustrates the fractions of pore-filling material (PPFM) and contact cement volume (FCC). In track (d), PPFM is shown as scatter points overlaid on the combined volume fractions of anhydrite and clay. Track (e) depicts the degree of sorting as scatter points, with the combined anhydrite and clay plotted as a continuous line. Track (f) showcases the multi-mineral distribution, with each mineral represented by a distinct color as indicated in the legend.

CRedit authorship contribution statement

Prasad Pothana: Writing – review & editing, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kegang Ling:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition. **Richard A. Schultz:** Writing – review & editing, Validation, Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Pearson's correlation coefficient

Pearson's correlation coefficient (r) is a measure of the linear correlation between two variables, x and y . It ranges from -1 to $+1$, where $+1$ indicates a perfect positive linear relationship, -1 indicates a perfect negative linear relationship, and 0 indicates no linear relationship. Pearson's correlation coefficient (r) can be expressed as

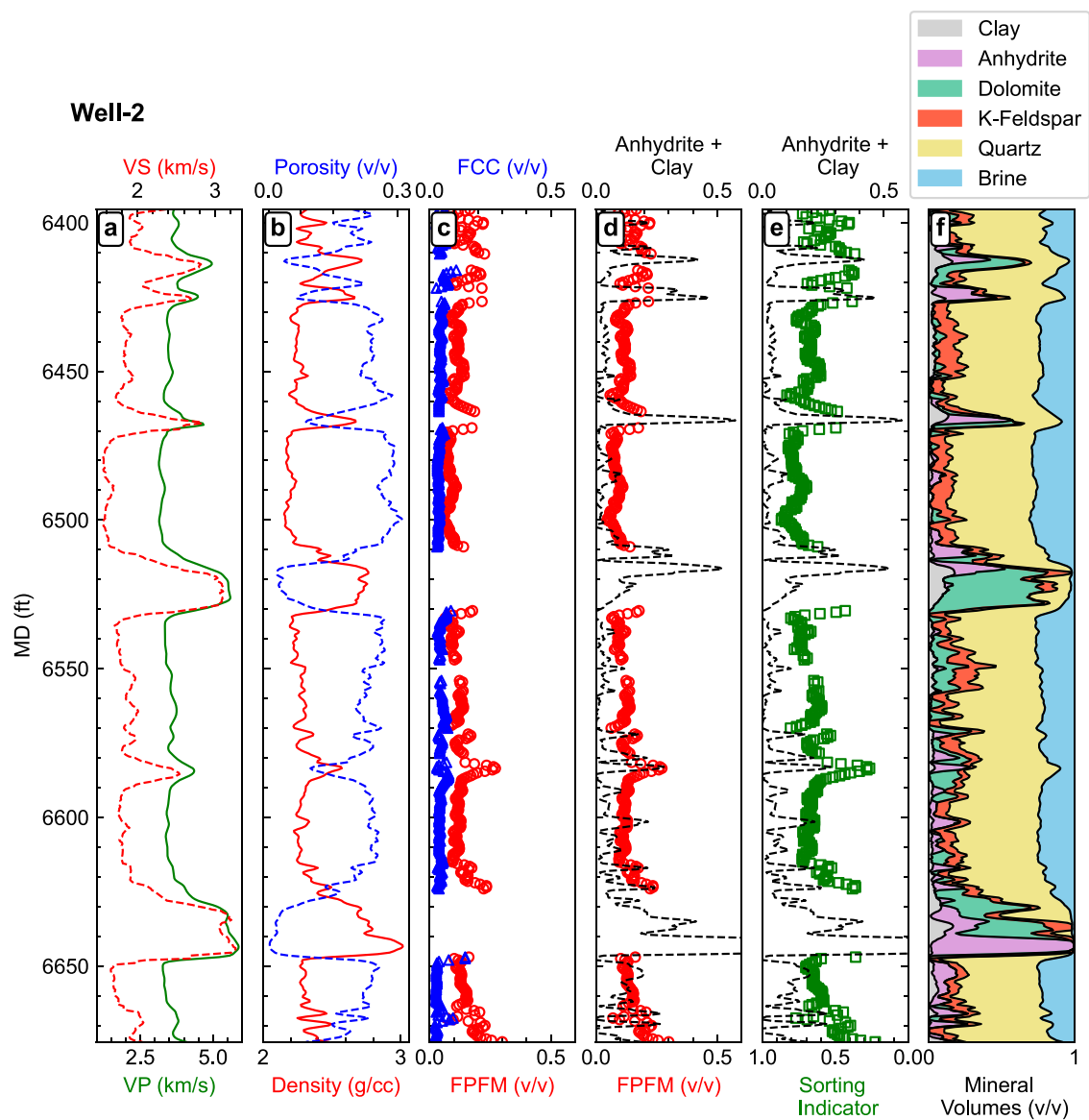


Fig. B.26. Composite log, rock physics modeling, and petrophysical analysis results for Well-2. Track (a) displays compressional and shear wave velocities. Individual track information is same as in Fig. B.25.

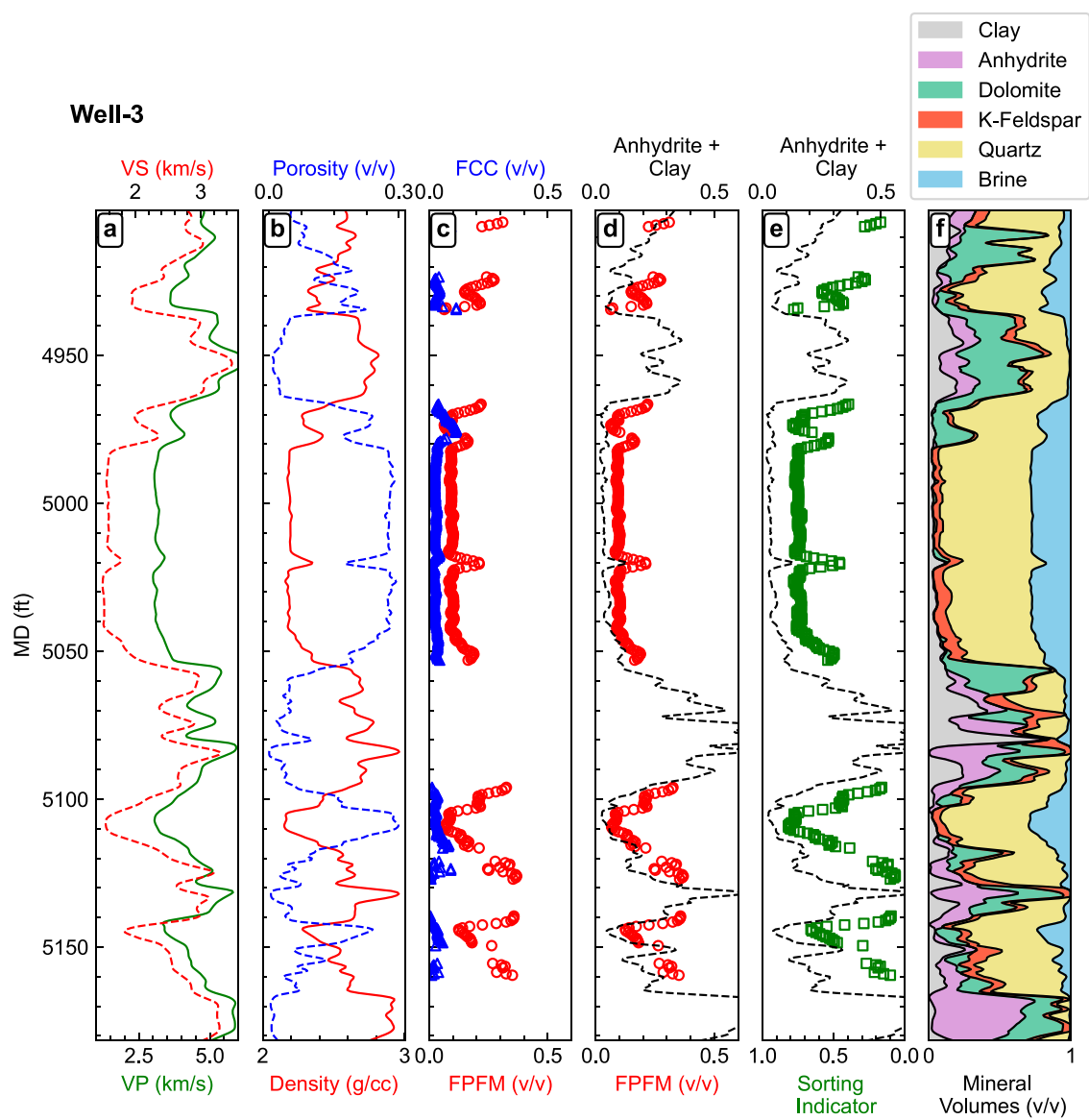


Fig. B.27. Composite log, rock physics modeling, and petrophysical analysis results for Well-3. Track (a) displays compressional and shear wave velocities (V_p and V_s). Individual track information is same as in Fig. B.25.

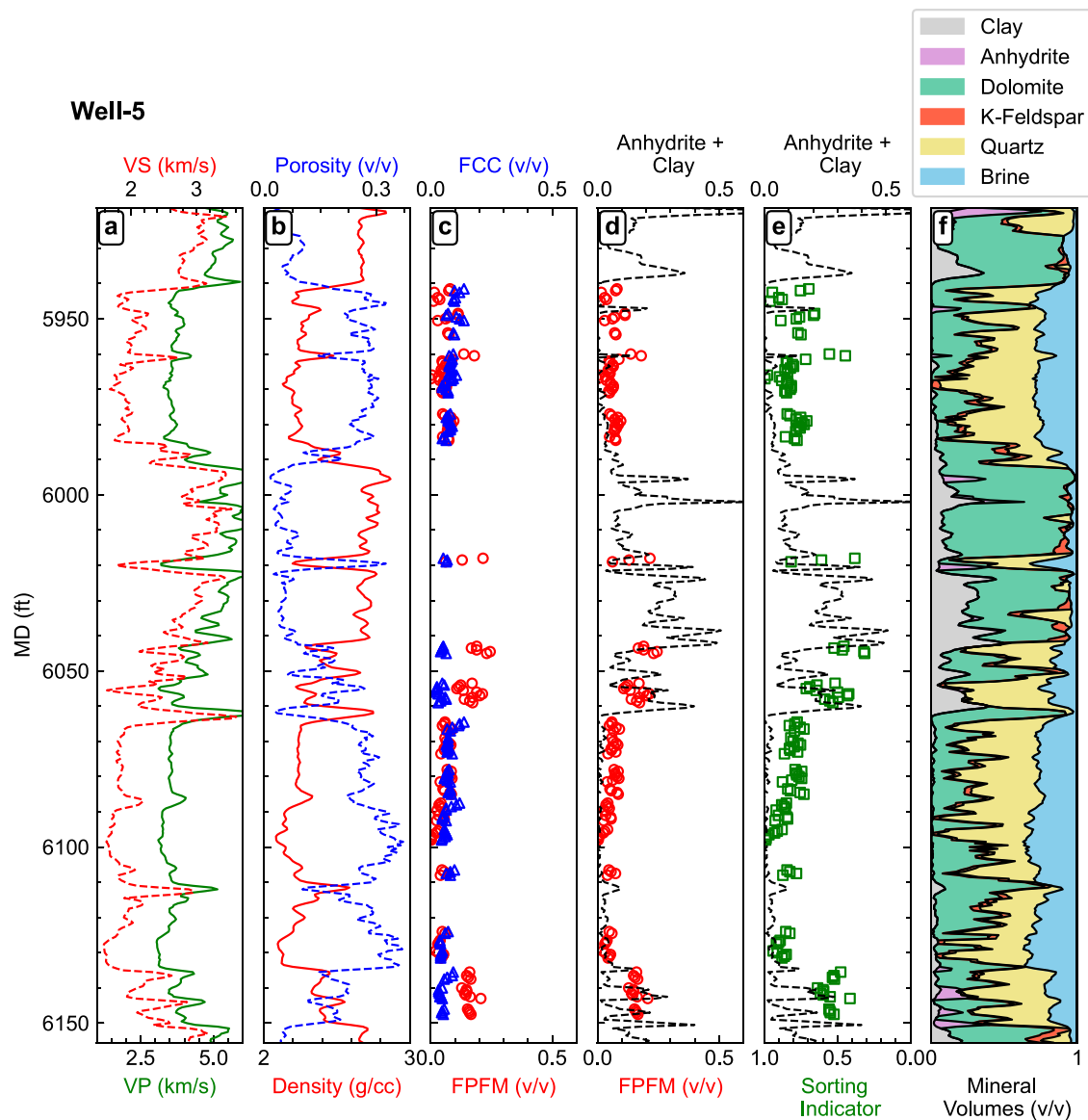


Fig. B.28. Composite log, rock physics modeling, and petrophysical analysis results for Well-5. Track (a) displays compressional and shear wave velocities. Individual track information is same as in Fig. B.25.

(e.g., Hahs-Vaughn (2023));

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \cdot \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (\text{A.1})$$

Appendix B. Petrophysical evaluation and rock physics modeling results

The composite logs for the five wells are shown in Figs. B.25, B.26, B.27 and B.28. The composite logs include the following tracks:

Data availability

The authors do not have permission to share data.

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